

Stress tests and model monoculture[☆]Keeyoung Rhee^{a,*}, Keshav Dogra^b^a Sungkyunkwan University (SKKU), 25-2 Seonggyungwan-ro, Jongno-gu, Seoul, 03063, Republic of Korea^b Federal Reserve Bank of New York, 33 Liberty Street, New York, NY, 10045, USA

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ABSTRACT

We study whether regulators should reveal stress test results that contain imperfect information about banks' financial health. Although disclosure restores market confidence in banks, it misclassifies some healthy banks as risky. This encourages banks to choose portfolios deemed safe by regulators, leading to *model monoculture* and making the financial system less diversified. Under the ex-ante optimal disclosure policy, the regulator addresses this tradeoff by fully revealing stress test results when adverse selection is very severe or very mild, but never disclosing the results otherwise.

1. Introduction

Stress tests have become a key tool for supervisors since the 2007–2009 Global Financial Crisis. By providing credible information about banks' financial health, stress tests can restore confidence in the banking system in a crisis; a testing regime can also limit the accumulation of risks ex ante. However, stress tests cannot perfectly predict how banks will react to all possible adverse shocks. First, stress tests inevitably ignore some sources of risk because the tests use only a few scenarios. Furthermore, stress tests may not even correctly predict how banks perform in the stress scenarios that do materialize. For example, the Office of Federal Housing Enterprise Oversight (OFHEO) conducted a risk-based capital stress test for the government-sponsored enterprises (GSEs) Fannie Mae and Freddie Mac before the subprime mortgage crisis. However, the OFHEO's stress test underestimated both the decline in house prices and the GSEs' exposure to this risk: realized defaults were 4–5 times greater than predicted by the model. Both GSEs were deemed insolvent in September 2008, even though the OFHEO's stress

test had misjudged them to be well-capitalized just six months earlier (Frame et al., 2015).

Due to this imperfect predictability, policymakers such as Bernanke (2013) and Hirtle (2018) raised the concern that routine stress testing and full disclosure of test results may lead to an unintended consequence called “*model monoculture*.” That is, banks may ignore their own (possibly superior) risk management systems; instead, the banks choose portfolios to game the regulators' models, leaving the whole system vulnerable to risks that the regulators' models do not anticipate. This concern highlights an important tradeoff associated with the disclosure of stress test results: disclosing (imperfect) test results can help restore confidence during a crisis but may also induce banks to disregard the risks not incorporated into the stress test scenarios, increasing systemic risk.

We use a stylized three-period model to analyze how regulators should optimally address this tradeoff when deciding whether to release stress test results. In the first period, “good” banks choose between two financial projects to invest in. Two different states of the world can realize in the third period, which we interpret as different stress scenarios.

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Each type of project yields a positive return in one state of the world, and zero in the other. We assume that different projects pay off in different states, reflecting the fact that different asset-liability profiles make a bank disproportionately exposed to different financial risks. A bank's choice of project can be interpreted as its portfolio; it can also be seen as representing which stress scenario the bank monitors more carefully and incorporates into its internal risk model.

In the second period, banks must borrow from competitive risk-averse lenders to continue their projects. However, banks may suffer credit rationing due to the presence of “bad” banks, who borrow from lenders like good banks, but whose projects yield nothing in any state. The fraction of these bad banks is a random variable that realizes before the financial market opens, representing the fact that the health of the overall financial system can fluctuate. Lenders do not observe a bank's project type, which creates an adverse selection problem: if the fraction of bad banks is high enough, lenders become unwilling to lend to any bank due to the fear of funding bad banks, increasing the banks' cost of funding. Social welfare is maximized by a diversified financial system, in which enough banks are solvent in every state that lenders' consumption does not vary too much across states.

To resolve the adverse selection problem, the regulator can conduct a stress test, identify how banks perform in a stress scenario, and reveal that information to the public. However, the regulator's information is imperfect; she can only test whether banks' projects pay off in one stress scenario, but cannot evaluate their solvency in the other scenario. As a result, the regulator can only verify that “approved” banks, whose projects pay off in the scenario incorporated in her stress test, are good. However, the other good banks, whose projects perform well in a different scenario not anticipated by the regulator, are “disfavored” by the stress test – this test cannot distinguish these banks from bad banks. We assume that before good banks choose their projects, the regulator commits to a disclosure policy contingent on the realized fraction of the bad banks (i.e., the severity of adverse selection), which specifies the probability that a randomly selected “approved” bank passes the test. Stress test results are revealed to lenders in the second period before the capital market is open, but after good banks choose their projects and the fraction of the bad banks is realized.

Our first finding is that disclosing stress test results reduces adverse selection problems, and always increases social welfare *ex post* – that is, taking banks' project choices as given. Disclosure of stress test results increases welfare through two channels. First, it eliminates adverse selection faced by “approved” banks by identifying them as good, making them fully funded. Second, it reduces socially wasteful lending to bad banks: lenders understand that banks that failed the stress test are likely to be the bad types, and therefore curtail lending to these failing banks. Thus, it is *ex-post* optimal to disclose stress test results. In addition, these *ex-post* welfare benefits of disclosure increase with the severity of adverse selection problems. Thus, our analysis captures the notion that the disclosure of supervisory information about bank solvency – such as the release of the bank-specific Supervisory Capital Assessment Program (SCAP) stress test results in May 2009 – can strengthen confidence in the financial system, especially during financial crises (Morgan et al., 2014).

Despite these benefits, full disclosure of stress test results makes the financial system more exposed to risks not captured by the regulator's model. Banks know that the only way to obtain cheap financing is to choose “approved” projects that pass the regulator's stress test. However, banks are reluctant to choose “disfavored” projects which perform well in the stress scenario not captured by the regulator's model: even though these “diversifying” projects would increase social welfare by keeping the banking system well-capitalized in both stress scenarios, banks choosing these projects will definitely fail the test and suffer higher borrowing costs. As a consequence, an inefficiently large fraction of banks will choose “approved” projects rather than “disfavored” ones, making the banking system less diversified compared to the economy absent disclosure. That is, a policy of full disclosure makes the

whole economy more vulnerable to the stress scenario neglected by the regulator's model, which we interpret as the model monoculture. Our analysis, therefore, has the empirical implication that stress test disclosure should increase the portfolio similarity of tested banks, consistent with Bräuning and Fillat (2019).

Given the tradeoff, we characterize the *ex-ante* optimal disclosure policy. Our main result is that the optimal policy is non-monotonic in the severity of adverse selection. Specifically, we find that the optimal policy features a bang-bang rule: the regulator should reveal all the “approved” banks when the adverse selection problem is either very severe or very mild, but should never disclose any test results otherwise. To understand why, first suppose adverse selection problems are severe enough that banks suffer credit rationing. In this case, revealing more “approved” banks has both an *ex-post* benefit (it alleviates adverse selection) and an *ex-ante* cost (it encourages good banks to underinvest in “disfavored” projects). Crucially though, the *marginal* welfare cost of disclosure is decreasing in the degree of disclosure. If the regulator discloses only a few “approved” banks, banks who are publicly *known* to be good will be extremely scarce, and will be able to borrow at very low interest rates – incentivizing good banks to choose “approved” projects. However, this incentive will diminish as the regulator discloses more “approved” banks, making them less scarce. Thus, partial disclosure is suboptimal: conditional on choosing disclosure, it is better to disclose *all* approved banks to minimize the scarcity premium earned by banks who pass the test. In particular, since the *ex-post* welfare benefit from disclosure is increasing in the severity of adverse selection, the regulator discloses all the “approved” banks when adverse selection is sufficiently severe.

Next, suppose adverse selection is mild so the good banks, regardless of their choice of project, are always fully funded. In this case, disclosure of stress test results does not have the *ex-post* benefit of alleviating adverse selection; it only influences good banks' *ex-ante* choice of project, and therefore, model monoculture. Surprisingly, disclosure can *improve* diversification when the fraction of bad banks is extremely low. To see why, recall that good banks are reluctant to choose “disfavored” projects due to the fear of a high cost of financing after failing the stress test. Lenders are then willing to lend to the “disfavored” banks on very favorable terms – if they can be identified as such – as a hedge against risks not captured by the regulator's model. The regulator can indirectly identify these “disfavored” banks by disclosing all the “approved” banks when there is a small fraction of bad banks; the remaining undisclosed banks are very likely to be “disfavored” types and enjoy a scarcity premium, partially restoring the incentive of good banks to choose the “disfavored” projects and thus reducing the model monoculture problem. However, zero disclosure is optimal when the fraction of bad banks is not sufficiently small; the undisclosed banks are then likely to be the bad ones, so the positive effect of disclosure on diversification disappears.

The remainder of this paper proceeds as follows. Section 2 presents our model. Section 3 characterizes the partial equilibrium at date 1. In Section 4, we characterize the full equilibrium at date 0 and find an *ex-ante* optimal disclosure policy. In Section 5 we extend our baseline model to discuss how our results depend on investors' risk aversion and on the extent to which stress tests are predictably biased. Section 6 reviews related work. Finally, Section 7 concludes the paper. All proofs are relegated to the Appendix.

2. Model

There are three dates: $t = 0, 1, 2$. At date 0, there is a continuum of “good” banks with measure 1 that undertake long-term financial projects that create stochastic returns at date 2. Moreover, a continuum of outside investors with measure 1 fund those projects at date 1 by purchasing bonds issued by the banks. Finally, a regulator, who seeks to maximize social welfare, decides whether to reveal information about banks at date 1, before outside investors purchase bonds. At

date $t = 2$, returns of the financial projects are realized and accrue to the banks and the outside investors.

At date 0, nature randomly draws a state $\omega \in \{G, D\}$, where $Pr(\omega = G) = p \in (0, 1)$; the two states represent different crisis scenarios. No one knows the true state until $t = 2$. After the underlying state of the economy is drawn, each “good” bank chooses one of two financial projects $\theta \in \{\gamma, \delta\}$, where project γ returns one unit of cash flow at $t = 2$ if $\omega = G$ and nothing if $\omega = D$, and project δ returns one unit of cash flow at $t = 2$ only in state $\omega = D$. As will become clear, state G represents a crisis scenario which is well-understood by the regulators and captured in their stress test models, while state D represents a scenario which is not. We prefer to think of the “project” as the bank’s portfolio: project γ represents a portfolio robust to the risks captured by the regulators’ models, while δ represents a portfolio vulnerable to these risks. One can also interpret the banks’ choice of project as a choice of risk model. That is, choosing project γ can be interpreted as choosing the same risk model as the regulator (and choosing a portfolio based on that model), while choosing δ can be interpreted as choosing a different risk model.

After the good banks choose their projects, a continuum of “bad” banks with measure $b \geq 0$ enters. Their project – henceforth called type β – produces zero output in every state, but yields a private benefit to each bad bank. Thus these banks always strictly prefer to continue their projects, if they can do so at zero cost.¹ The measure of these bad banks $b \geq 0$ is a random variable with a continuous distribution function $F(\cdot)$ over the support $[0, \infty)$. This captures the idea that adverse selection problems may be more or less severe at different times and allows us to ask how disclosure policy should optimally adjust to high- and low-stress episodes. Flannery et al. (2013) document empirically that bank opacity, or the accuracy with which outsiders can assess a bank’s value, varies substantially over time and that it increased substantially during the 2007 - 2009 Global Financial Crisis. We call each bank’s financial project $\theta \in \{\gamma, \delta, \beta\}$ selected at $t = 0$ its “type.”

At $t = 1$, banks – including the bad ones – must raise $x \in (0, 1)$ dollars to continue their financial projects. They raise these funds by issuing bonds to a continuum of outside investors with measure one. We assume

$$x < \min\{p, 1 - p\}, \quad (1)$$

which guarantees that investing in any good project $\theta \in \{\gamma, \delta\}$ has a positive net present value. The bidding game is described in detail in Online Appendix B. In brief, the bonds issued by the banks are sold as follows. Each investor $i \in [0, 1]$ places a buy order (q_i, R_i) such that the investor offers to purchase q_i units of bonds for repayment terms R_i per unit of capital. Whenever possible, trade takes place at a market-clearing interest rate R such that the supply of bonds issued by banks equals the amount demanded by investors requesting a rate of return R or lower. At this market-clearing interest rate, each bank is matched with a representative sample of investors, who lend to the banks at the rates promised. However, if the total supply of bonds exceeds the demand at the highest interest rate R_i offered by investors, banks are stochastically rationed such that only some fraction of banks are matched with investors. Finally, at date 2, each bond-holder receives R_i per bond issued by a solvent bank, i.e., a bank whose payoffs from its project exceed its liabilities. If a bank’s liabilities exceed the project’s return, we assume for simplicity a 100% bankruptcy cost, i.e., the bank’s bond-holders receive nothing. Online Appendix B shows that all investors place the same buy order, i.e., $(q_i, R_i) = (q, R)$ for all $i \in [0, 1]$ in any equilibrium.

Every investor has preferences represented by

$$U(c_1, c_2) = c_1 + p \log c_G + (1 - p) \log c_D. \quad (2)$$

¹ When we study optimal policy, we will assume the private benefit accruing to bad banks is not added to the regulator’s social welfare function. The reader is welcome to assume that the bad banks’ private benefit converges to zero.

where c_1 is consumption at date 1, and c_G and c_D denote consumption at date 2 in states G and D , respectively. In Section 5.1, we show that our results are not qualitatively changed if investors have CRRA utility at $t = 1$ with risk aversion greater than 1. Note that since the investors are risk averse, they would ideally like to hold bonds issued by both type γ and δ banks to smooth consumption across states G and D at date 2. Each investor is endowed with a sufficiently large amount of cash to fund all the banks’ projects at date 1. For analytical simplicity, we further assume that each investor owns an equal amount of shares issued by every bank. Under this assumption, all residual returns of the banks after repaying the bonds are equally distributed to the investors; thus, social welfare is simply calculated as the aggregated expected utility of the investors. All the good banks, whether type γ or δ , behave in the best interest of their equity holders (i.e., the investors). Finally, each bank’s type is private information, which creates an adverse selection problem.

A regulator maximizes social welfare by deciding whether to reveal her superior but imperfect information about the banks’ private types before the capital market opens at date 1. Specifically, the regulator can conduct a stress test to determine whether each bank’s project pays off in state G (a crisis scenario which is well-understood by the regulator and captured in its models), but cannot determine whether the bank’s project pays off in state D (which is not well-understood). All banks are subject to the regulator’s stress test. Thus, after conducting the stress test, the regulator can distinguish type γ banks from the others, but cannot distinguish between type δ and β banks. Formally, we assume that for each individual bank indexed by $k \in [0, 1 + b]$, the regulator has access to a signal $\tau_k \in \{0, 1\}$ about bank k ’s type $\theta_k \in \{\gamma, \delta, \beta\}$ with the following conditional probabilities:

$$Pr(\tau_k = 1 | \theta_k = \gamma) = 1, \quad Pr(\tau_k = 1 | \theta_k = \delta) = 0, \quad Pr(\tau_k = 1 | \theta_k = \beta) = 0. \quad (3)$$

We assume that the regulator can choose the probability α with which she discloses type γ banks. Formally, the regulator chooses a disclosure policy $\alpha \in [0, 1]$ such that if bank k ’s type is γ , with probability α the regulator reveals her information by sending the message $\sigma_k = 1$, and with probability $1 - \alpha$ the regulator does not disclose her information, i.e., she reports $\sigma_k = 0$. Furthermore, if bank k ’s type is either δ or β , the regulator has no information on bank k ’s type so reveals nothing by reporting $\sigma_k = 0$. The regulator receives this information before the banks enter the capital market. Hence, if the regulator discloses her information on bank k ’s type θ_k , the investors update their belief about θ_k and place their buy orders accordingly. Throughout, we abbreviate θ_k, τ_k and σ_k to θ, τ and σ , respectively, since all banks with the same type behave symmetrically in equilibrium, as we will show later. We occasionally refer to the banks whose types are not revealed by the regulator (i.e., $\sigma = 0$) as the ones “failing” the stress test, and refer to type γ banks whose type is revealed by the regulator (i.e., $\sigma = 1$) as “passing” the stress test.

To study how different policy regimes affect diversification in the financial system, we assume that the regulator can commit to a disclosure policy $\alpha : [0, \infty) \rightarrow [0, 1]$ before banks choose their types at $t = 0$, where $\alpha(b)$ is the probability that a type γ bank is disclosed given that the measure of bad banks is b . A policy in which the regulator always discloses her information ($\alpha(b) = 1$ for all $b > 0$) is analogous to routinized stress tests, such as the Dodd-Frank Act Stress Tests (DFAST) and the Comprehensive Capital Analysis and Review (CCAR) for large-sized U.S. financial firms. Conversely, the stress tests conducted contingently in the event of financial crises can be thought of as a policy of revealing information only when adverse selection in the financial market is very severe (i.e., b is high). Our assumption that the regulator observes the fraction of bad banks (but not which specific banks are bad) captures the idea that policymakers have some information about the severity of adverse selection and the funding pressures facing banks. Therefore, it is feasible to follow a policy of disclosing stress test results only when overall adverse selection problems are especially severe. Indeed, the Board of Governors took the unusual step of revealing information on

bank-level losses under the 2009 SCAP precisely to reduce uncertainty and restore confidence in the banking system during the financial crisis (Board of Governors of the Federal Reserve System, 2009).

Our stylized model abstracts from a number of features of real-world stress tests for ease of exposition. First, in reality, banks are directly penalized for failing stress tests: for example, under CCAR, the Fed forbids failing banks from making dividend payments or stock repurchases. We abstract from these direct consequences, to focus on the indirect costs that arise because investors are less willing to lend to failing banks. Incorporating direct costs would only strengthen our main result that disclosure can reduce diversification in the financial system.

Another simplifying assumption is that some banks fail in every state of the world, and the regulator's information concerns whether a bank's project will pay off unconditionally. In reality, banks generally only fail in a crisis, and thus stress tests concern the probability of banks' capital shortfall conditional on a crisis. To model this explicitly, we could introduce a shock $s \in \{0, 1\}$ realized at date 1, where $s = 0$ represents "normal times" and $s = 1$ represents a rare, adverse event. If $s = 0$, all projects pay off 1 for sure and banks do not require funding to continue their projects. If $s = 1$, the model proceeds as described earlier, and projects may not pay off at date 2 depending on the value of $\omega \in \{G, D\}$. Clearly, maximizing the ex-ante bank profits (respectively, social welfare) will be equivalent to maximizing profits (respectively, social welfare) in the adverse state. It is thus without loss of generality to assume that the adverse scenario always realizes.² This simplifies the model and restricts our attention to outcomes in a crisis scenario, which are relevant to stress tests.

Lastly, we assume that the total measure of the good banks is fixed but the measure of the bad banks is random. One may argue that a more plausible source of variations in the severity of adverse selection problems would be fluctuations in the *proportion* of bad banks, keeping the total measure fixed. We do not follow this approach in our model, because, as we describe in Online Appendix C, such an alternative framework turns out to be less tractable while delivering qualitatively similar results only in a restricted setting of the investors' risk appetite.

3. Ex-post partial equilibrium at date 1

3.1. Equilibrium characterization

We now describe equilibrium outcomes in the capital market at date 1 ($t = 1$) after good banks choose their types, the fraction of type β "bad" banks $b \geq 0$ is realized, and the regulator chooses the fraction $\alpha \in [0, 1]$ of type γ banks to be disclosed. At $t = 1$, there are two different groups of banks in the capital market: a group of banks who "pass the test" ($\sigma = 1$), and the remaining banks who "fail the test" ($\sigma = 0$). Outside investors (henceforth investors) then evaluate the default risks of the banks in the two groups differently; these risks depend on both the disclosure policy $\alpha \in [0, 1]$ and the fraction of type β banks. The group labeled $\sigma = 1$ consists of only type γ banks and has measure αg : every bank in this $\sigma = 1$ group repays its debt in state G , but defaults in the other state D . In the $\sigma = 0$ group, there are type γ banks with measure $(1 - \alpha)g$, type δ banks with measure $(1 - g)$, and type β banks with measure b . Thus the investors expect that a bank in the $\sigma = 0$ group repays its debt in the state G with probability $\frac{(1 - \alpha)g}{1 - \alpha g + b}$ and in the state D with probability $\frac{1 - g}{1 - \alpha g + b}$, respectively.

Since the investors have different beliefs about the default risks of the banks in the two groups, the regulator's disclosure policy effectively

segments the date-1 capital market into two submarkets. In the submarket for the banks with $\sigma = 1$, the investors make a symmetric buy order (q_1, R_1) for the bonds issued by these banks, and a fraction $\phi_1 := \frac{q_1}{\alpha g}$ of the $\sigma = 1$ banks receive funding. Similarly, in the submarket for the banks labeled $\sigma = 0$, the investors make a symmetric buy order (q_0, R_0) , and a fraction $\phi_0 := \frac{q_0}{1 - \alpha g + b}$ of the banks borrow from the investors at the market. For expositional convenience, rather than expressing the buy orders in terms of the quantity of bonds purchased (together with the interest rate), i.e., (q_0, R_0) and (q_1, R_1) , we hereafter express them in terms of the fraction of banks receiving funding, i.e., as (ϕ_0, R_0) and (ϕ_1, R_1) , respectively.

We next discuss the date-1 expected utility function of each investor as a prospective debt holder. In state G , each investor, as an equity holder, will receive the dividend y_G , an equal share of the net income that every type γ bank earns. Furthermore, each investor will receive a promised debt repayment made by type γ banks who obtained funding at $t = 1$. A measure αg of type γ banks were labeled $\sigma = 1$, and investors choose the fraction ϕ_1 of these banks to fund at gross interest rate R_1 ; the remaining type γ banks with measure $(1 - \alpha)g$ were labeled $\sigma = 0$, and investors fund the fraction ϕ_0 of these banks at rate R_0 . Thus, the total debt repayment each investor receives in state G equals $\alpha g \phi_1 R_1 x + (1 - \alpha)g \phi_0 R_0 x$. Similarly, in state D , each investor will get the dividend y_D that every type δ bank equally pays to all equity holders. Furthermore, type δ banks fulfill the debt repayment $(1 - g)\phi_0 R_0 x$. Lastly, each investor spends $(\alpha g \phi_1 + (1 - \alpha g + b)\phi_0)x$ to purchase bonds at $t = 1$. In sum, the investors' utility function in (2) can be rewritten as

$$p \log(y_G + \alpha g \phi_1 R_1 x + (1 - \alpha)g \phi_0 R_0 x) + (1 - p) \log(y_D + (1 - g)\phi_0 R_0 x - (\alpha g \phi_1 + (1 - \alpha g + b)\phi_0)x).$$

When choosing how much debt to purchase at date 1, each investor takes y_G and y_D as given, but chooses her purchase orders (ϕ_0, R_0) and (ϕ_1, R_1) to maximize her expected utility. Then, the buy orders (ϕ_0, R_0) and (ϕ_1, R_1) are an equilibrium strategy if and only if the investor's first order conditions are satisfied — i.e., the marginal benefit of purchasing ϕ_σ units of bonds from the banks in group $\sigma \in \{0, 1\}$ equals the marginal cost of funding. Taking partial derivatives with respect to ϕ_0 and ϕ_1 , we have the following first order conditions:

$$\begin{aligned} \frac{p}{c_G} R_1 &= 1, \\ \left[\left(\frac{(1 - \alpha)g}{1 - \alpha g + b} \right) \frac{p}{c_G} + \left(\frac{1 - g}{1 - \alpha g + b} \right) \frac{(1 - p)}{c_D} + \left(\frac{b}{1 - \alpha g + b} \right) \times 0 \right] R_0 &= 1. \end{aligned} \quad (4) \quad (5)$$

The conditions (4) and (5) can be interpreted as standard Euler equations: the equilibrium interest rates depend on the probability that the banks will repay in the respective states of the world and the investors' marginal utility of funding. First, (4) indicates that any bank in the submarket $\sigma = 1$ is type γ with probability 1, and surely repays R_1 in state G with the risk-weighted probability $\frac{p}{c_G}$. Next, (5) indicates that each bank in the submarket $\sigma = 0$ is either type γ with probability $\frac{(1 - \alpha)g}{1 - \alpha g + b}$, type δ with probability $\frac{1 - g}{1 - \alpha g + b}$, or type β with probability $\frac{b}{1 - \alpha g + b}$. If the bank is type γ , it repays R_0 in state G with the risk-weighted probability $\frac{p}{c_G}$. If the bank is type δ , it repays R_0 in state D with the risk-weighted probability $\frac{1 - p}{c_D}$. The bank defaults on its debt in any state if its type is β .

Lastly, we characterize equilibrium strategies. In equilibrium, all financial returns are symmetrically distributed to investors as either dividends or debt repayments. Hence, we must have $c_G = g(\alpha \phi_1 + (1 - \alpha)\phi_0)$ and $c_D = (1 - g)\phi_0$. By plugging these formulas into (4) and (5), we can derive the equilibrium buy orders (ϕ_0^*, R_0^*) and (ϕ_1^*, R_1^*) as follows.

² Another possibility is that the regulator can disclose information in "normal times," before the adverse scenario is realized. In this case, it may no longer be true that maximizing ex-ante profits or welfare is equivalent to maximizing profits or welfare in the crisis state.

Lemma 1. Suppose the regulator discloses type γ banks with probability α . Then, there exists a unique partial equilibrium in which each investor places the buy orders (ϕ_0^*, R_0^*) in the submarket $\sigma = 0$ and (ϕ_1^*, R_1^*) in the submarket $\sigma = 1$, where:

- (i) $(\phi_0^*, R_0^*) = \left(1, \frac{(1-\alpha)g + (1-g)b}{(1-\alpha)p + (1-p)}\right)$ if $(p - gx)\alpha + xb \leq 1 - x$;
- (ii) $\phi_0^* < 1$ and $R_0^* = \frac{1}{x}$ if $(p - gx)\alpha + xb > 1 - x$, where ϕ_0^* is implicitly determined by

$$p \frac{1}{\frac{\alpha}{1-\alpha} + \phi_0^*} + (1-p) \frac{1}{\phi_0^*} = ((1-\alpha)g + (1-g)b)x. \quad (6)$$
- (iii) $(\phi_1^*, R_1^*) = \left(1, \frac{(\alpha + (1-\alpha)\phi_0^*)g}{p}\right)$ for every $\alpha > 0$ and $b \geq 0$.

Proof. See Appendix A.1. $\square$

To help understand the equilibrium strategies, we first discuss the $\sigma = 0$ submarket. Investors' willingness to lend to the banks depends on the degree of adverse selection in the capital market ($b \geq 0$). As shown by (5), a higher value of b reduces the marginal benefit of purchasing an extra unit of bonds because it increases the risk that an investor funds an unproductive type β bank. In order for the investors to be willing to fully finance the banks in the $\sigma = 0$ submarket (i.e., $\phi_0^* = 1$) despite this risk, the banks must add a risk premium to the bond rate R_0^* . This risk premium increases as adverse selection becomes more severe (i.e., b increases), as can be seen in Lemma 1-(i), which states that R_0^* is increasing in b .

However, R_0^* cannot exceed $\frac{1}{x}$: since a bank requires x dollars to continue its project that yields at most one dollar, it would default on a loan with $R_0 > \frac{1}{x}$ for sure, and the investors would refuse to lend at such a high rate. Thus, the banks cannot fully compensate the investors for their wasteful lending to type β banks when b is sufficiently high. In this case, every penny of the bank's financial return is spent repaying bondholders (i.e., $R_0^* = \frac{1}{x}$), and the investors optimally respond to the insufficient risk premium by reducing their lending to the $\sigma = 0$ banks to avoid funding type β banks (i.e., $\phi_0^* < 1$). That is, the $\sigma = 0$ banks are rationed in the capital market when the adverse selection problem is severe (Lemma 1-(ii)). Moreover, the fraction of banks rationed in the $\sigma = 0$ submarket increases with b because the investors feel less compensated for the risks of wasteful lending as more banks in the $\sigma = 0$ submarket are type β . Lastly, for a given b , more banks in the $\sigma = 0$ submarket are rationed as α increases, i.e., the regulator discloses more information to the investors. As more type γ banks are revealed to pass the stress test, and are moved from the $\sigma = 0$ to the $\sigma = 1$ group, the average quality of the remaining banks in the $\sigma = 0$ pool declines (i.e., such a bank is more likely to be of type β).

In contrast, the banks in the $\sigma = 1$ submarket are fully financed since investors know that none of these banks are type β , and all debts of these banks will be repaid in state G (Lemma 1-(iii)). Interestingly, the interest rate R_1^* is (weakly) decreasing in b since ϕ_0^* (weakly) decreases with b . Namely, the investors offer a discount to the bond rate to the banks in the $\sigma = 1$ submarket since these banks are less likely to be of type β than the banks in the $\sigma = 0$ submarket.

Lastly, we discuss the equilibrium in a *laissez-faire* economy where the regulator does not disclose any test result (i.e., $\alpha = 0$), so every bank is effectively in the $\sigma = 0$ (sub-)market. Indeed, this equilibrium is a limiting case of the equilibrium described by Lemma 1.

Corollary 1. Suppose the regulator does not disclose any test results to the market (i.e., $\alpha = 0$). Then there is a unique equilibrium in which each investor places the symmetric buy order (ϕ^*, R^*) in the capital market, where:

- (i) If $b \leq \frac{1}{x} - 1$, $(\phi^*, R^*) = (1, 1 + b)$;

- (ii) If $b > \frac{1}{x} - 1$, $(\phi^*, R^*) = \left(\frac{1}{x(1+b)}, \frac{1}{x}\right)$.

It is worth noting that $\phi_1^* \geq \phi_0^* \geq \phi_0^*$ for every $b \geq 0$ and $\alpha \in [0, 1]$. That is, when some banks are labeled $\sigma = 1$ (i.e., are revealed to pass the test), this worsens the rationing faced by a "good" bank labeled $\sigma = 0$, relative to a counterfactual in which no test results are disclosed. In particular, when b is sufficiently low – specifically, $\left(\frac{1}{x} - 1\right) - \alpha \left(\frac{p}{x} - g\right) < b \leq \frac{1}{x} - 1$ – good banks labeled $\sigma = 0$ are rationed, whereas they would not be rationed at all in the *laissez-faire* economy. One might then wonder whether disclosure might reduce the efficiency of capital allocation by worsening credit rationing for some "good" banks. The next section formally analyzes the ex-post welfare effects of disclosing stress test results.

3.2. The ex-post optimal disclosure policy

We next analyze how disclosure policy influences ex-post welfare, which is measured by the investors' $t = 1$ expected utility. By plugging (ϕ_0^*, ϕ_1^*) as described in Lemma 1 into the investors' utility function, we can write ex-post welfare as the following function of (g, α, b) , denoted by $U(g, \alpha, b)$:

$$U(g, \alpha, b) = p \log g + (1-p) \log(1-g) + [p \log(\alpha + (1-\alpha)\phi_0^*) - gx(\alpha + (1-\alpha)\phi_0^*)] + [(1-p) \log \phi_0^* - (1-g)\phi_0^*x] - b\phi_0^*x. \quad (7)$$

The first term represents the welfare benefit from a diversified financial system, which is maximized at $g = p$. Since the investors are risk averse, it is socially optimal for some good banks to invest in different projects which pay off in different states of the world. The second term indicates the net social benefit from lending to type γ banks; lending increases the date-2 utility in state G , but consumes resources and reduces utility at date 1. Analogously, the third term represents the net gains from lending to type δ banks. Finally, the last term is the loss from lending to type β banks. The ex-post welfare without any disclosure ($\alpha = 0$) is then $U(g, 0, b)$, where

$$U(g, 0, b) = [p \log g + (1-p) \log(1-g)] + p \log \phi^* + (1-p) \log \phi^* - (1+b)\phi^*x.$$

We can thus define the total ex-post welfare gains from a disclosure policy $\alpha > 0$, relative to no disclosure, as $U(g, \alpha, b) - U(g, 0, b)$. It is worth noting that $U(g, \alpha, b) - U(g, 0, b) = 0$ whenever $\phi_1^* = \phi^* = \phi_0^* = 1$ – i.e., when b is sufficiently low that no bank is ever rationed, either with or without disclosure. Then we can analyze the welfare effect of the disclosure policy $\alpha > 0$ as the following theorem states. Throughout, we let $x \wedge y$ and $x \vee y$ denote $\min\{x, y\}$ and $\max\{x, y\}$, respectively.

Lemma 2.

- (i) For a fixed $\alpha > 0$, $U(g, \alpha, b) - U(g, 0, b) = 0$ if $b \leq \left(\frac{1}{x} - 1\right) - \alpha \left(\frac{p}{x} - g\right)$, but $U(g, \alpha, b) - U(g, 0, b) > 0$ and $U(g, \alpha, b) - U(g, 0, b)$ is increasing in b otherwise.
- (ii) For a fixed $b \geq 0$, $U(g, \alpha, b) - U(g, 0, b) = 0$ if $\alpha \leq \left[0 \vee \left(\frac{1-x(1+b)}{p-gx}\right)\right] \wedge 1$, but $U(g, \alpha, b) - U(g, 0, b) > 0$ and $U(g, \alpha, b) - U(g, 0, b)$ is increasing in α otherwise.

Proof. See Appendix A.1. $\square$

Note from Lemma 2-(i) that disclosure only increases welfare when adverse selection in the *laissez-faire* economy is extremely severe. In other words, the disclosure will not affect the allocation of capital if

every bank is fully financed in the capital market with or without the regulator's information, which is the case when adverse selection is relatively mild (i.e., b is small). Likewise, disclosure cannot affect welfare unless the regulator identifies a large enough fraction of type γ banks (i.e., α is sufficiently high) that banks labeled $\sigma = 0$ are rationed.

For sufficiently high b and α , releasing the stress test results increases welfare in two ways. First, disclosure (partially) eliminates the adverse selection problem faced by type γ banks. Specifically, each type γ bank, with probability α , is labeled $\sigma = 1$ and is therefore fully financed in the capital market; such a bank labeled $\sigma = 1$ is perfectly distinguished from type- β "lemon" banks labeled $\sigma = 0$. Second, releasing the stress test results reduces socially wasteful lending to type β banks. That is, investors increase their subjective probability that a bank labeled $\sigma = 0$ is type β (relative to the probability they would assign in the *laissez-faire* economy) and therefore curtail their lending to this bank. Obviously, such credit rationing inefficiently reduces productive lending to the good banks labeled $\sigma = 0$. Nevertheless, Lemma 2 reveals that the positive effect of the credit rationing in the $\sigma = 0$ submarket outweighs the negative effect. This positive welfare effect is larger when the capital market suffers a more severe adverse selection problem (i.e., b is high); it is more socially useful to ration credit to type β banks when there are more of these banks.

Lemma 2-(ii) also implies that full disclosure ($\alpha = 1$) is an ex-post optimal policy. For any given disclosure policy $\alpha \in [0, 1]$, the regulator can always improve welfare by releasing more information on banks' private types, thus alleviating the remaining adverse selection problem in the capital market. Increasing α means that more type γ banks are correctly identified as good, so they are fully financed in the capital market; it also increases rationing for type β banks by reducing the average quality of the $\sigma = 0$ pool, thus reducing wasteful lending.

4. Ex-ante equilibrium and optimal disclosure policy

We have shown that disclosing stress test results can improve social welfare by alleviating ex-post adverse selection in the capital market. However, the disclosure also affects the relative return to γ and δ projects and thus influences good banks' project choice *ex ante*. Good banks will find project δ less attractive when it is expected to suffer severe adverse selection problems relative to project γ , encouraging them to choose project γ instead. As a result, disclosure may lead to *under-diversification* in the overall economy; there will be too many type γ banks and too few type δ banks, leaving investors with lower consumption in state D relative to state G . Thus, when designing an optimal disclosure policy, the regulator should consider its effects on aggregate portfolio diversification.

In this section, we analyze how the regulator's *disclosure policy* $\alpha(b)$ influences each bank's ex-ante choice of project. Specifically, before banks choose their types $\theta \in \{\gamma, \delta\}$, the regulator commits to the fraction $\alpha(b)$ of type γ banks to be disclosed for each possible realization of $b \geq 0$, the degree of adverse selection, at date 1. We investigate how each disclosure policy $\alpha(b)_{b \geq 0}$ contributes to (in-)efficiency in diversification. Then we find the optimal disclosure policy given this tradeoff between ex-post adverse selection and ex-ante diversification.

We first describe how banks choose their types $\theta \in \{\gamma, \delta\}$ at date 0. Recall that every investor owns an equal share – as an *equity holder* – of each bank. Furthermore, there is no moral hazard between good banks and their equity holders: each bank aims to maximize the expected utility of its equity holders. Given a particular realization of $b \geq 0$ and a disclosure policy $\alpha \in [0, 1]$, $\omega_\gamma = \alpha(1 - R_1^*x) + (1 - \alpha)\phi_0^*(1 - R_0^*x)$ is the dividend paid by each type γ bank in state G , and $\omega_\delta = \phi_0^*(1 - R_0^*x)$ is the dividend paid by each δ bank in state D (recall that $\omega_d = 0$ when δ banks are rationed and spend all the revenues from their project on interest payments, i.e., $R_0^* = \frac{1}{x}$). Furthermore, each investor will also receive payments at date 2 on the bonds they purchased from banks at date 1. Specifically, $z_G := \alpha g R_1^*x + (1 - \alpha)g\phi_0^* R_0^*x$ is the debt repayment from type γ banks in state G , $z_D := (1 - g)\phi_0^* R_0^*x$ is the debt

repayment from type δ banks in state D , and $C = \alpha g x + (1 - \alpha g + b)\phi_0^*x$ is the total investment made at the date-1 capital market. Hence, the expected utility of each outside investor – which is the objective function for each bank's date-0 maximization problem – is

$$p \log c_G + (1 - p) \log c_D - C = p \log(\omega_\gamma g + z_G) + (1 - p) \log(\omega_\delta(1 - g) + z_D) - C.$$

In equilibrium, g is determined to maximize the expected utility of outside investors across all realizations of b , taking $\omega_\gamma, \omega_\delta, z_G, z_D$ and C as given. If a marginal bank chooses project γ rather than δ , each equity holder's utility in state G will increase by the dividend ω_γ multiplied by $\frac{1}{c_G}$, the marginal utility in state G . However, the investor's utility in state D will decrease by the dividend ω_δ weighted by the marginal utility $\frac{1}{c_D}$. We throughout define $\Delta(g, \alpha, b)$ as the *net* gains to the outside investors from an additional type γ bank, given the fraction g of type γ banks, a realization of b , and a disclosure policy α . Formally, $\Delta(g, \alpha, b)$ is defined as

$$\Delta(g, \alpha, b) := \frac{p}{c_G} \omega_\gamma - \frac{1 - p}{c_D} \omega_\delta.$$

Indeed, $\Delta(g, \alpha, b)$ is the difference between the expected marginal gains and losses from having a marginal type δ bank switch its project to type γ ; equivalently, it is the partial derivative of expected utility with respect to g (we provide an explicit expression for $\Delta(g, \alpha, b)$ in Appendix A.2).

Since social welfare is defined as the investors' expected utility, the regulator's welfare maximization problem is

$$\max_{\{\alpha(b) \in [0, 1] : b \in [0, \infty)\}} \int_0^\infty U(g, \alpha(b), b) dF(b), \quad (8)$$

subject to the following constraint, reflecting banks' optimal choice of project:

$$\int_0^\infty \Delta(g, \alpha(b), b) dF(b) = 0. \quad (9)$$

The constraint (9) states that in any equilibrium, every individual bank must be indifferent between choosing projects γ and δ *ex ante*. For example, if $\int_0^\infty \Delta(g, \alpha(b), b) dF(b) > 0$, the expected marginal benefit from an extra type γ bank is strictly positive, making project γ a strictly more attractive choice than project δ to the banks. It is worth noting that $\Delta(g, \alpha, b)$ is decreasing in g : the marginal benefit of another type γ bank decreases with the population of type γ banks. Intuitively, a high fraction of type γ banks means large consumption of equity holders in state G relative to state D , while it lowers the marginal value of having another type γ bank. This implies that the investors as equity holders, like the regulator, value diversification.³

Crucially, any change in disclosure policies $\alpha(\cdot)$ shifts $\Delta(g, \alpha(b), b)$, and therefore influences overall diversification (i.e., the value of g that satisfies (9)). For example, disclosing test results for high values of b pools type δ banks together with a large number of type β banks, increasing type δ banks' borrowing costs and thus reducing their gross dividend payouts ω_δ . Such a policy thus encourages good banks to choose project γ rather than δ , formulated by an increase in $\int_0^\infty \Delta(g, \alpha(b), b) dF(b)$ for a given g . As a result, more good banks strategically respond by choosing project γ rather than δ , increasing g to an extent that the constraint (9) continues to be binding. In this context,

³ In particular, we cannot have a corner solution in which all banks strictly prefer the same project – either $g = 1$ and $\int_0^\infty \Delta(g, \alpha(b), b) dF(b) > 0$, or $g = 0$ and $\int_0^\infty \Delta(g, \alpha(b), b) dF(b) < 0$. This follows from the assumption that investors have log utility for their date-2 consumption, which implies $\lim_{c \rightarrow 0} \frac{d}{dc} \log c = \infty$.

the equilibrium condition (9) summarizes how the disclosure of “biased” stress test results affects diversification.

As a benchmark, it is useful to consider an unconstrained program in which the regulator maximizes (8) ignoring the constraint (9), i.e., she can simply dictate that measure $g \in (0, 1)$ of banks choose type γ , whatever her choice of disclosure policy $\alpha(\cdot)$ is. In this relaxed optimization problem, the regulator would choose $g = p$ since full diversification is always socially desirable. Moreover, we know from Lemma 2 that full disclosure always weakly maximizes ex-post social welfare. Therefore, an unconstrained-optimal disclosure policy would be $\alpha(b) = 1$ for every $b \geq 0$. The presence of the constraint (9), however, makes such an allocation unimplementable. Indeed, the following observation reveals that disclosure of the regulator’s information always reduces diversification in the financial system.

Lemma 3. *If a disclosure policy $\alpha(\cdot)$ yields $g = p$, then we must have $\alpha(b) = 0$ with probability 1. If $\alpha(b) > 0$ with positive probability, then we must have $g > p$. In particular, any optimal disclosure policy must yield $g > p$.*

Proof. See Appendix A.2. $\square$

The regulator’s stress tests are biased against type δ banks: they correctly identify type γ banks as “good,” but do not distinguish between type δ banks and type β “bad” banks. Disclosure of such biased stress test results inevitably reduces diversification ($g > p$) relative to the first-best outcome ($g = p$). Thus, our model has the empirical implication that disclosure of stress test results should cause banks to choose more similar portfolios, consistent with Bräuning and Fillat (2019), who find that stress-tested banks became more likely to have similar loan portfolios after the implementation of DFAST in 2011, while this trend was not present for non-stress-tested banks. (In particular, they find that banks with poor stress test results adjust their portfolios to become similar to banks which performed well.) In our model, it is *feasible* to keep diversification at its first-best level, but this can only be implemented by completely refraining from disclosure (i.e., $\alpha(b) = 0$ with probability 1). Such a policy is strictly suboptimal because the regulator forgoes an opportunity to alleviate the ex-post adverse selection. Instead, a constrained-efficient policy balances the benefit and cost of disclosing stress test results. In what follows, we describe in more detail the structure of optimal disclosure policies which manage this trade-off arising from the inherent bias in the regulator’s stress test model. Throughout, we denote an ex-ante optimal disclosure policy by $\alpha^{**}(b)$ for every $b \geq 0$.

4.1. Optimal disclosure under mild adverse selection

We begin by characterizing the optimal disclosure policy $\alpha(\cdot)$ for low values of $b \leq \frac{1-p}{x} - (1-g)$. When b is sufficiently low, banks will never be rationed in the date-1 capital market even if they are labeled as “not good” (i.e., $\sigma = 0$). That is, we have $\phi_0^* = \phi_1^* = \phi^* = 1$ no matter what the level of disclosure $\alpha \in [0, 1]$ is. Consequently, the degree of disclosure in these states does not directly affect ex-post social welfare — i.e., $U(g, \alpha, b) = U(g, 0, b)$ for every $\alpha > 0$. Hence, the optimal disclosure policy $\alpha(b)$ under sufficiently mild adverse selection should be designed to minimize the (negative) impact of the stress tests on diversification captured by $\Delta(g, \alpha, b)$. By Lemma 3, any optimal disclosure policy yields under-diversification ($g > p$). Since $\Delta(g, \alpha, b)$ is decreasing in g , the regulator’s objective is to minimize $\int_0^\infty \Delta(g, \alpha(b), b) dF(b)$ in order to raise $(1-g)$, the measure of relatively scarce type δ banks. The following lemma describes how an optimal disclosure policy can accomplish this goal.

Lemma 4. *There exists a $\underline{b}^{**} := \frac{g}{p} - 1$ such that any optimal disclosure policy $\alpha^{**}(\cdot)$ must have $\alpha^{**}(b) = 1$ for all $b \leq \underline{b}^{**}$ and $\alpha(b) = 0$ for all $b \in (\underline{b}^{**}, \frac{1-p}{x} - (1-g)]$.*

Proof. See Appendix A.2. $\square$

Underlying Lemma 4 is the fact that $\Delta(g, \alpha, b)$ is strictly decreasing in α if $b < \underline{b}^{**}$ but strictly increasing if $b \geq \underline{b}^{**}$. Hence, the regulator can optimally mitigate the under-diversification problem by following a cut-off rule: full disclosure ($\alpha(b) = 1$) when the adverse selection problem is sufficiently mild (i.e., $b < \underline{b}^{**}$), and no disclosure ($\alpha(b) = 0$) otherwise.

The result that full disclosure for sufficiently low b can *improve* diversification might seem paradoxical. To understand this, first recall from Lemma 3 that disclosure reduces the supply of bonds issued by type δ banks relative to its first-best level (i.e., $1-g < 1-p$). Since bonds issued by type δ banks are scarce in the capital market, the investors have low consumption in state D , and are willing to pay a very high price for the bonds that are surely issued by type δ banks. Thus, if the regulator could somehow identify type δ banks to the investors, these banks will enjoy a *scarcity* premium when borrowing in the capital market at date 1.

Of course, the regulator cannot *directly* identify type δ banks: she cannot distinguish between type δ banks (that the investors would like to lend to) and type β banks (that investors wish to avoid lending). However, when the fraction of bad banks b is sufficiently small, one way to partially identify type δ banks is to reveal all of the type γ banks to investors. There are few type β banks when b is very low, so the remaining banks who are *not* identified as type γ are likely to be “good” type δ banks rather than “bad” type β banks. Hence, full disclosure maximizes the likelihood that these “undisclosed” banks are type δ , which maximizes the scarcity premium that type δ banks can enjoy. From an ex-ante perspective, this improvement in the expected funding conditions of type δ banks encourages good banks to choose project δ at date 0 *ex ante*.

However, this “paradoxically” favorable effect of full disclosure vanishes as b no longer has a sufficiently low value. As b becomes larger, a bank known to be *not* type γ is quite likely to be a type β bank. In this case, disclosing all (or even some) of the type γ banks would worsen the ex-post adverse selection and thus increase the cost of funding for the “not-type- γ ” banks — in particular, type δ banks. As a result, good banks are incentivized to choose project type δ at date 0. Thus, for relatively high values of b , the regulator should adopt a zero disclosure policy to minimize the adverse effect on diversification.

4.2. Optimal disclosure under severe adverse selection

Next, we characterize optimal disclosure policy when $b > \frac{1-p}{x} - (1-g)$, i.e., adverse selection is relatively severe. In this case, credit rationing may take place in the $\sigma = 0$ submarket. To characterize the structure of the optimal disclosure policy, it is useful to work with the Lagrangian of the regulator’s optimization problem (8) subject to (9), which is expressed as

$$\mathcal{L} = \int [U(g, \alpha(b), b) - \lambda \Delta(g, \alpha(b), b)] dF(b).$$

The Lagrangian multiplier λ is strictly positive: given log utility over date-2 consumption, the constraint (9) is always binding for some interior values of $g \in (0, 1)$. For each b , we next define the net welfare gains from a disclosure policy $\alpha > 0$ as follows:

$$W(g, \alpha, b) := [U(g, \alpha, b) - \lambda \Delta(g, \alpha, b)] - [U(g, 0, b) - \lambda \Delta(g, 0, b)]. \quad (10)$$

For a given b , a disclosure policy α increases welfare ($W(g, \alpha, b) > 0$) if and only if the positive effect on the alleviation of ex-post adverse selection ($U(g, \alpha, b) - U(g, 0, b)$) is large relative to the negative impact on ex-ante diversification ($\Delta(g, \alpha, b) - \Delta(g, 0, b)$). If a disclosure policy $\alpha^{**}(\cdot)$ is optimal, we must have $W(g, \alpha^{**}(b), b) > 0$ for any b such that $\alpha^{**}(b) > 0$.

In our model, the net benefit from revealing stress test results varies continuously with α , i.e., the proportion of type γ banks whose type is revealed to the investors. However, this does not necessarily mean

that an optimal disclosure policy $\alpha^{**}(b)$ has an interior value for every degree of adverse selection b . Instead – surprisingly – the optimal disclosure policy exhibits a bang-bang rule.

Lemma 5. *There exists a cutoff value $\bar{b}^{**} \geq \frac{1-p}{x} - (1-g)$ such that $\alpha^{**}(b) = 0$ (i.e., the regulator discloses no type γ banks) if $\frac{1-p}{x} - (1-g) < b \leq \bar{b}^{**}$ and $\alpha^{**}(b) = 1$ (i.e., the regulator discloses all type γ banks) if $b > \bar{b}^{**}$.*

Proof. See Appendix A.2. $\square$

Lemma 5 rests on two crucial features. First, “partial disclosure,” i.e., revealing type γ banks with a probability α strictly between 0 and 1, is strictly suboptimal when adverse selection is severe. To understand why, first consider a disclosure policy α that is positive but so close to zero that even the banks “failing” the stress test do not suffer credit rationing (i.e., $\phi_0^* = \phi_1^* = 1$). In this case, disclosure does not directly increase social welfare by rationing credit to type β banks (i.e., $U(g, \alpha, b) = U(g, 0, b)$), but does increase good banks’ ex-ante incentive to select project γ . On the other hand, even a small $\alpha > 0$ worsens diversification because it raises $\Delta(g, \alpha, b)$ above $\Delta(g, 0, b)$, further incentivizing good banks to choose project γ . Since such a small $\alpha > 0$ does not increase welfare and worsens diversification, it cannot be optimal.⁴

Next, consider an intermediate value of α that causes type δ banks to suffer credit rationing at the $t = 1$ capital market (i.e., $\phi_0^* < 1$) but is still far below unity. In this case, while disclosure does directly increase social welfare, it only does so by a small amount, while it has a large adverse effect on diversification. Recall that when adverse selection is severe, bonds issued by good banks (of either type γ or δ) are relatively scarce. Thus, if the regulator reveals only a small fraction of type γ banks, their bonds – claims on a bank which is known to be “not-of-type- β ” – are extremely scarce in the capital market so the investors are willing to purchase these bonds even at a very high price. Since passing the stress test becomes very valuable, good banks are strongly discouraged from choosing project type δ (which gives zero chance of passing the stress test), significantly lowering diversification. On the other hand, we know from Lemma 2 that disclosing a small fraction α of type γ banks does not substantively alleviate ex-post adverse selection and thus creates little direct welfare benefit. In sum, disclosing a relatively small fraction of type γ banks yields the net welfare loss and thus cannot be optimal.

However, the marginal welfare benefit from revealing more type γ banks can outweigh the cost incurred by under-diversification when α is very large. If the regulator reveals a large fraction of type γ banks, their bonds are not very scarce in the capital market any longer. That is, the borrowing costs of these revealed type γ banks are not substantially low. This makes the benefit from passing the stress test – while still positive – smaller than it would be for relatively low values of α , mitigating the adverse effect of disclosure on diversification. At the same time, Lemma 2 implies that the ex-post welfare benefit from disclosure is relatively large when α is close to 1. Crucially, for α close to 1, the marginal benefit of increasing α is always positive, as the marginal adverse effect on diversification is falling in α , whereas the direct welfare benefit on adverse selection is increasing. Thus, once the regulator discloses her information, it is always optimal to choose full disclosure (i.e., $\alpha = 1$).

⁴ In the proof Lemma 5 in Appendix A.2, we formally find a threshold $\hat{\alpha} \geq 0$ such that $\phi_0^* = 1$ for every $\alpha \leq \hat{\alpha}$. Then the discussion above in the text can be alternatively expressed as follows: any $\alpha \in (0, \hat{\alpha}]$ is strictly suboptimal for each $b > \frac{1-p}{x} - (1-g)$. In Appendix A.2, we further show that $\hat{\alpha}$ is monotone decreasing in b . Intuitively, as the adverse selection problem grows more severe, even a disclosure policy revealing only a small fraction of type γ banks can improve ex-post social welfare.

Given that it is optimal to disclose either fully or not at all, for which realizations of b is full disclosure optimal? The second key feature underlying Lemma 5, which helps answer this question, is that the net welfare benefit from full disclosure increases as adverse selection becomes more severe — i.e., $W(g, 1, b)$ is increasing in b . To see this, recall from (10) that

$$W(g, 1, b) = [U(g, 1, b) - U(g, 0, b)] - \lambda[\Delta(g, 1, b) - \Delta(g, 0, b)],$$

where the first term indicates the direct welfare benefit from full disclosure and the second term represents the welfare loss from full disclosure arising from under-diversification. In Lemma 2, we have already seen that the first term is strictly increasing in b for every $b > \frac{1-p}{x} - (1-g)$ — that is, the direct welfare benefit of full disclosure increases when the adverse selection problem becomes more severe.

We next show that the second term representing the adverse effect of full disclosure on diversification is monotone decreasing in b . First, by definition of $\Delta(g, \alpha, b)$, we have $\Delta(g, 1, b) = \frac{p}{g} - x > 0$ (which is independent of b) for every $b > \frac{1-p}{x} - (1-g)$. When b exceeds the threshold $\frac{1-p}{x} - (1-g)$, full disclosure always leads to credit rationing for type δ banks (i.e., $R_0^* = \frac{1}{x}$), implying that these type δ banks cannot pay out any dividend in state D no matter how far above the threshold b is. Thus, under full disclosure, banks’ ex-ante incentive to choose project γ no longer depends on the severity of adverse selection once the adverse selection problem is sufficiently severe.

Next, by plugging $\alpha = 0$ into $\Delta(g, \alpha, b)$, we have

$$\Delta(g, 0, b) = \left(\frac{p}{g} - \frac{1-p}{1-g} \right) \max\{1 - R^*x, 0\}$$

Recall from Lemma 3 that there is always under-diversification ($p < g$) in equilibrium under any optimal policy, making type δ banks scarce in the capital market relative to type γ banks. Furthermore, any bank, regardless of its type, faces the same funding condition in $t = 1$ capital market under no disclosure. As a result, equity holders find type δ banks more valuable than type γ banks, which is expressed by $\Delta(g, 0, b) \leq 0$.

However, $\Delta(g, 0, b)$ is weakly increasing in b ($\frac{d}{db}\Delta(g, 0, b) \geq 0$), i.e., the magnitude of this “scarcity premium” that equity holders attach to type δ banks is decreasing in b . This follows from Corollary 1: the gross interest rate R^* repaid to bondholders weakly increases with the degree of adverse selection. Intuitively, type δ banks are more valuable than γ banks only when these banks have sufficiently large net income after repaying their debts at date 2. The higher the degree of adverse selection b , the higher the interest rate, and the smaller the dividend a type δ bank pays in state D (as well as the dividend a type γ bank pays in state G). Thus, we have $\frac{d}{db}\Delta(g, 0, b) \geq 0$: the benefit of an additional type δ bank eventually vanishes as the adverse selection problem worsens.

Consequently, even though full disclosure of stress test results always *somehow* increases banks’ incentive to choose project γ when $b > \frac{1-p}{x} - (1-g)$, this negative effect on diversification weakens as b increases. Indeed, type δ banks would have been rationed in the capital market and thus pay zero dividends even absent disclosure. In other words, the adverse effect of full disclosure on diversification is weakly decreasing in the severity of adverse selection. Putting these all together, we have that $W(g, 1, b)$ is strictly increasing in b , which implies the optimal disclosure policy must have the single-crossing property for $b > \frac{1-p}{x} - (1-g)$ as stated in Lemma 5: the regulator optimally chooses full disclosure if b exceeds a threshold $\bar{b}^{**} \geq \frac{1-p}{x} - (1-g)$, but zero disclosure otherwise.

By combining Lemma 4 and 5, we can fully characterize the structure of the optimal disclosure policy.

Theorem 1. *An optimal disclosure policy $\alpha^{**}(\cdot)$ is characterized by two cutoffs $0 < \underline{b}^{**} < \bar{b}^{**} < \infty$ such that $\alpha^{**}(b) = 1$ if $b \in [0, \underline{b}^{**}] \cup [\bar{b}^{**}, \infty)$ and $\alpha^{**}(b) = 0$ otherwise. Any optimal disclosure policy admits an excessive supply of bonds issued by type γ banks, i.e., $g > p$.*

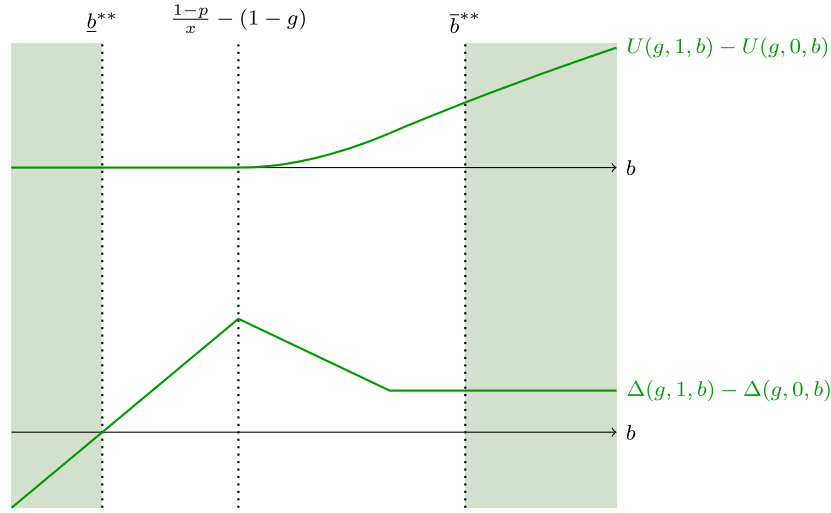


Fig. 1. The structure of the ex-ante optimal disclosure policy.

Proof. See Appendix A.2. $\square$

Fig. 1 graphically illustrates how optimal policy is shaped by two cutoffs. The lower threshold b^{**} is the lowest value of b such that $\Delta(g, 1, b) - \Delta(g, 0, b) \leq 0$. As discussed in Section 4.1, when $b < b^{**}$, disclosure of the regulator's information increases welfare by improving diversification because banks identified as either type δ or β are very likely to be δ , and therefore disclosure allows type δ banks to obtain cheaper funding. For intermediate values of b ($b^{**} < b \leq \frac{1-p}{x} - (1-g)$), disclosure worsens diversification ($\Delta(g, 1, b) - \Delta(g, 0, b) > 0$), whereas it does not address adverse selection problems at all ($U(g, 1, b) - U(g, 0, b) = 0$). Hence, the regulator should refrain from disclosing any information.

For high values of b ($b > \frac{1-p}{x} - (1-g)$), it can be seen that disclosure alleviates adverse selection problems ($U(g, 1, b) - U(g, 0, b) > 0$), and this welfare benefit increases with the severity of adverse selection ($\frac{d}{db}[U(g, 1, b) - U(g, 0, b)] > 0$). Furthermore, it can be seen that $\Delta(g, 1, b) - \Delta(g, 0, b) > 0$ for every $b > \frac{1-p}{x} - (1-g)$: disclosure is costly due to the associated under-diversification. However, this cost of disclosure decreases with the severity of adverse selection ($\frac{d}{db}[\Delta(g, 1, b) - \Delta(g, 0, b)] < 0$). Hence, the net welfare gain from disclosure increases with the severity of adverse selection. Therefore, to maximize welfare, the regulator should disclose her information if and only if adverse selection problems are sufficiently severe, i.e., $b > \bar{b}^{**}$.

Another interpretation of the optimal disclosure policy is that the regulator releases her information when it is relatively accurate.⁵ To see this, note from (3) that the regulator's posterior belief about an individual bank receiving the signal $\sigma = 0$ is

$$Pr(\theta = \delta | \tau = 0) = \frac{1-g}{1-g+b}.$$

Since $Pr(\theta = \delta | \tau = 0) \rightarrow 1$ as $b \rightarrow 0$ and $Pr(\theta = \beta | \tau = 0) \rightarrow 0$ as $b \rightarrow \infty$, the regulator's signal becomes more accurate as b converges to either zero or infinity. Information disclosure can be costly when it is inaccurate, mistakenly classifying type δ banks as bad banks. To maximize welfare, the regulator only releases her information when it accurately identifies a bank's type, i.e., when b is either very low or very high.

Throughout, we have assumed that the private and social benefits from diversification coincide. As we have shown, even when the outside investors and equity holders fully internalize the social benefits

from a diversified financial system, disclosing imperfect stress test results can inefficiently reduce diversification if banks are able to game the stress tests. This cost would be even more severe to the extent that private individuals do not fully internalize the benefits from diversification: this would only amplify our main result. Indeed, a key justification of stress tests (and financial regulation more broadly) in practice is that in a crisis, low levels of bank capital have negative effects on the wider economy, which are not internalized by individual banks. To the extent that such externalities are present, policy should try even harder to induce banks to diversify their portfolios. Absent other instruments, this can only be achieved by releasing stress test results even less frequently.⁶

Finally, one should not necessarily conclude from our analysis that releasing stress test results always reduces diversification. Information disclosure is only costly to the extent that banks can anticipate which projects the regulator will deem as safe, which is a maintained assumption in the model. If instead banks do not know at date 0 whether type γ or δ project will "fail" at date 1, committing to unconditional disclosure may not reduce diversification, and would be possibly optimal. (In Section 5.2, we formalize this idea and numerically analyze an optimal level of unpredictability in the regulator's stress test models.) Thus, our model would suggest that regulators should attempt to conceal their models from subject banks, if possible. Recently, the Federal Reserve has discussed whether to disclose additional details about its supervisory models; some commentators have argued that the models should be fully disclosed.⁷ Our analysis lends support to concerns regarding the disclosure of supervisory risk models: increasing predictability of the supervisory stress testing could allow banks to manipulate their stress test results without prudently reducing actual risks, leading to model monoculture. Our analysis also suggests the empirical prediction that disclosing additional details about the models used in supervisory stress tests would cause an even larger increase in portfolio similarity among tested banks.

⁶ Earlier versions of this paper modeled this formally by studying an extension in which when banks' projects fail, they are forced to sell productive assets to repay depositors. This introduces a pecuniary externality, because an individual bank does not internalize that its asset sales reduce the prices that other banks receive for their assets. Asset sales reduce welfare because banks are more productive managers of assets. In this case, the optimal level of diversification is higher than in the model without externalities (the optimal g is lower).

⁷ For more details, see <https://www.gpo.gov/fdsys/pkg/FR-2017-12-15/pdf/2017-26856.pdf>.

⁵ The authors specially thank Youngwoo Koh for this helpful comment.

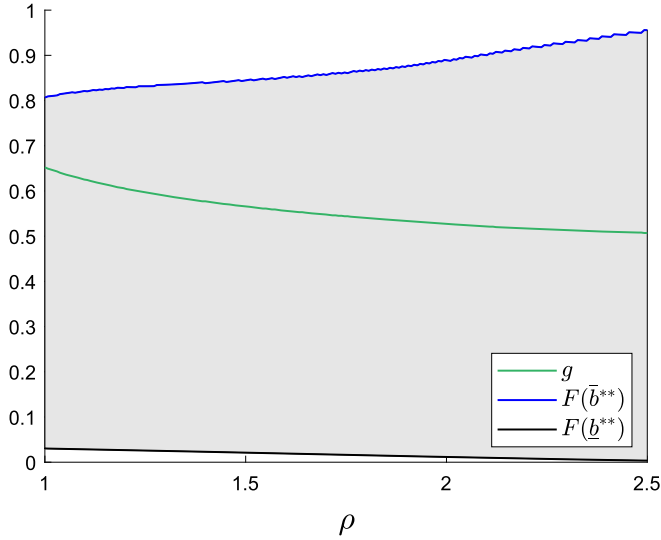


Fig. 2. Comparative statics of the optimal disclosure policy with respect to ρ .

5. Extensions

Our baseline model makes a number of simplifying assumptions for tractability. We now investigate whether (and how) relaxing these assumptions changes our main results.

5.1. More general CRRA preferences

Disclosure can be costly because it leads to an inefficiently under-diversified financial system. In our model, under-diversification is inefficient because investors are risk-averse. It is therefore natural to ask how the optimal policy changes as the investors become more risk-averse, and the value of diversification increases. To answer this question, we modify our main model by assuming that investors' preferences over their date-2 consumption are described by a CRRA utility function $u(c) = \frac{c^{1-\rho}}{1-\rho}$ with $\rho > 1$. In Appendix A.3, we show that the optimal disclosure policy has the same structure as that with $\rho = 1$ (i.e., $B^{**} := [0, \underline{b}^{**}] \cup [\bar{b}^{**}, \infty)$ for some $0 \leq \underline{b}^{**} \leq \bar{b}^{**} < \infty$). Furthermore, as in our baseline model, the optimal disclosure policy features a larger fraction of type γ banks than in the unconstrained efficient allocation.

Theorem 2. *When investors have CRRA preferences with $\rho > 1$, optimal disclosure policy $\alpha^{**}(\cdot)$ is characterized by two cutoffs $0 < \underline{b}^{**} < \bar{b}^{**}$ such that $\alpha^{**}(b) = 1$ if $b \in [0, \underline{b}^{**}] \cup [\bar{b}^{**}, \infty)$ and $\alpha^{**}(b) = 0$ otherwise. Any optimal disclosure policy admits excessive supply of bonds issued by type γ banks, i.e., $g^\rho > p$.*

Proof. See Appendix A.3. $\square$

To see how the optimal disclosure policy changes with the social value of diversification, we conduct a comparative statics exercise with respect to the degree of the investors' risk aversion $\rho > 1$. Specifically, we consider a numerical example in which $x = 0.2$, $p = 0.5$, and b – the population of type β banks on the market – is uniformly distributed over $[0, 10]$. Fig. 2 graphically illustrates how the optimal policy varies with ρ .

There are a couple of interesting observations. First, the equilibrium fraction g of type γ banks induced by the optimal disclosure policy converges downward to the first-best level $g = p^{\frac{1}{\rho}}$ as ρ increases. In part, this reflects investors' private incentives: as equity holders become more risk averse, banks will choose a more diversified mix of projects, for

the same disclosure policy, in order to maximize the equity holders' expected utility. In addition, the regulator takes into account the increased social value of diversification and discloses stress test results less often as the investors become more risk-averse, implementing an even higher level of portfolio diversification than would arise given the same disclosure policy. Indeed, Fig. 2 shows that the total probability of disclosure $F(\bar{b}^{**}) + (1 - F(\underline{b}^{**}))$ is decreasing in ρ . Specifically, one can see that the upper threshold $\bar{b}^{**}$ increases with ρ , whereas the lower threshold $\underline{b}^{**}$ decreases with ρ , increasing the size of the non-disclosure region (shaded gray).

5.2. Predictable bias of supervisory stress testing

In the last paragraph of Section 4, we remarked that model monoculture arises from the *predictability* of the bias in supervisory stress test models. Indeed, subject banks are able to game the system because these banks know precisely which type of risks the regulator focuses on in its supervisory stress tests. We also conjectured that full disclosure of stress test results would maximize welfare if banks were unable to predict the bias of the regulator's stress test models.

In this section, we extend our baseline model to prove our conjecture. Suppose that the regulator tests for project γ with probability μ (that is, distinguishes between γ on the one hand and δ and β on the other hand, as in our baseline model), while the regulator tests for project δ with probability $1 - \mu$. To simplify the analysis, we throughout assume $p = \frac{1}{2}$. If the regulator can costlessly choose any value of μ , it will be obviously optimal to set $\mu = \frac{1}{2}$ and $\alpha(b) = 1$ for all b . Under $\mu = \frac{1}{2}$, we have $\Delta(\frac{1}{2}, \alpha, b) = 0$ for all α, b , so full diversification ($g = \frac{1}{2}$) will obtain for any disclosure policy $\alpha(b)$.

Suppose then that $\mu \in (\frac{1}{2}, 1)$ is exogenous. With probability μ , the regulator learns the identities of all type γ banks (but not type δ banks), and can decide whether or not to release that information, as in our baseline model. Recall that one interpretation is that the regulator is able to study how banks will perform in a “stress scenario” corresponding to state G , but cannot study another risk corresponding to state D . The regulator then commits to a binary disclosure policy $\alpha_\gamma(b) \in \{0, 1\}$: $\alpha_\gamma(b) = 1$ if and only if the regulator reveals the test results under scenario G when the fraction of bad banks equals b . Similarly, with probability $1 - \mu$, the regulator learns the identities of all type δ banks, but not type γ banks (i.e., she can study a scenario based on state D , but not state G), and then commits to a binary disclosure policy $\alpha_\delta(b) \in \{0, 1\}$. That is, the disclosure policy is contingent on the scenarios that the regulator tests for.

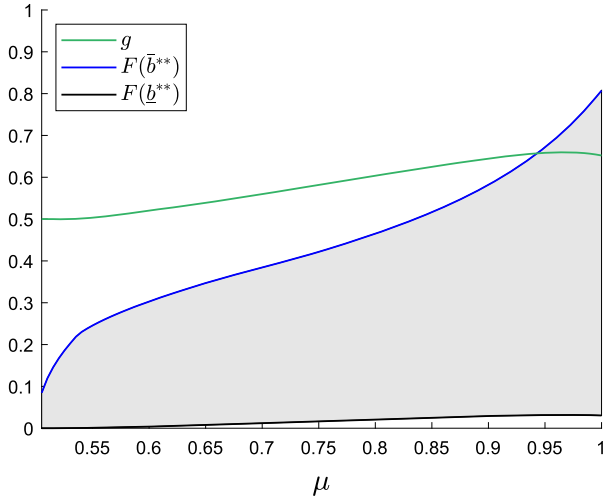
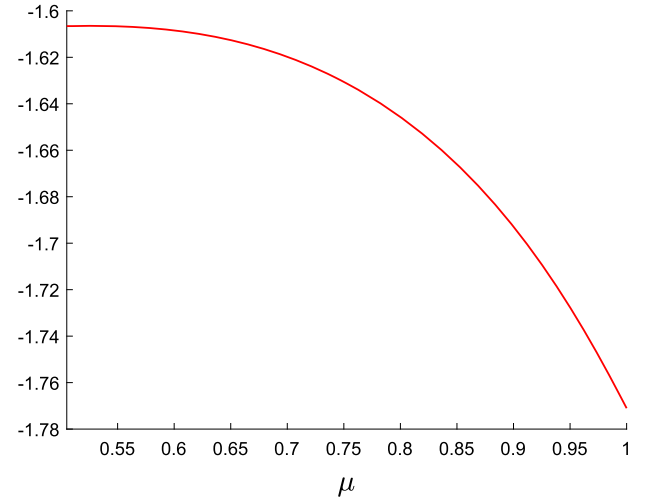
Given this uncertainty about which stress scenario the regulator will be able to test for, her optimization problem becomes:

$$\max_{g, \alpha_\gamma(\cdot), \alpha_\delta(\cdot)} \mu \int [\alpha_\gamma(b)U_\gamma(g, b) + (1 - \alpha_\gamma(b))U_n(g, b)]dF(b) \\ + (1 - \mu) \int [\alpha_\delta(b)U_\delta(g, b) + (1 - \alpha_\delta(b))U_n(g, b)]dF(b),$$

subject to

$$\mu \int [\alpha_\gamma(b)\Delta_\gamma(g, b) + (1 - \alpha_\gamma(b))\Delta_n(g, b)]dF(b) \\ + (1 - \mu) \int [\alpha_\delta(b)\Delta_\delta(g, b) + (1 - \alpha_\delta(b))\Delta_n(g, b)]dF(b) = 0.$$

Here $U_n(g, b) := U(g, 0, b)$ and $\Delta_n(g, b) := \Delta(g, 0, b)$ denote welfare and the net benefit to equity holders from project γ when the regulator does not reveal any information, respectively. Similarly, $U_\gamma(g, b) := U(g, 1, b)$ and $\Delta_\gamma(g, b) := \Delta(g, 1, b)$ denote welfare and the net benefit to equity holders from project γ when the regulator discloses the result of a test based on scenario G , respectively. These functions are described by the same expressions as in our baseline model with $\mu = 1$. Furthermore, $U_\delta(g, b)$ and $\Delta_\delta(g, b)$ denote welfare and the net benefit to equity

(a) Optimal policy ($\alpha_\gamma^{**}(\cdot)$)

(b) Welfare

Fig. 3. Comparative statics of the optimal disclosure policy with respect to μ .

holders from project γ when the regulator discloses the result of a test based on scenario D . These functions can be written as

$$U_\delta = p \log g + (1-p) \log(1-g) + p \log \phi_\delta - (1-g + \phi_\delta(g+b))x,$$

$$\Delta_\delta(g, b) = \frac{p}{g} \max \left\{ 1 - \frac{g+b}{p}x, 0 \right\} - \frac{1-p}{1-g} \left(1 - \frac{1-g}{1-p}x \right),$$

where $\phi_\delta = \min \left\{ \frac{p}{(g+b)x}, 1 \right\}$ denotes the fraction of γ and β banks who are rationed when the results of a test based on scenario D are disclosed.

The possibility of testing for project δ makes it difficult to analytically find an optimal disclosure policy ($\alpha_\gamma^{**}(\cdot), \alpha_\delta^{**}(\cdot)$) in general. Nevertheless, we can numerically find the optimal policy for each $\mu \in (\frac{1}{2}, 1)$ for a given distribution function $F(b)$. For example, Fig. 3 depicts the numerical results with $x = 0.2$ and $b \sim U[0, 10]$. (We stress that the qualitative results depicted here generally hold for $x < p = \frac{1}{2}$ and $b \sim U[0, \bar{b}]$ with $\bar{b} > 1$.) The left panel depicts the optimal disclosure policy when the regulator can test for scenario G , $\alpha_\gamma^{**}(\cdot)$, as a correspondence of $\mu \in (\frac{1}{2}, 1)$. When the regulator can test for scenario D , disclosure is always optimal ($\alpha_\delta^{**}(b) = 1$ for every $b \geq 0$), so we do not plot this policy. In addition, the right panel illustrates how welfare under the optimal policy varies with μ . Higher values of μ mean that the regulator's tests are more predictably biased towards type γ banks. In this context, our baseline model is viewed as the most predictable case at the rightmost point $\mu = 1$.

This numerical example provides some features worth discussing. First, since the regulator's model is, albeit stochastically, biased against state D ($\mu > \frac{1}{2}$), the optimal disclosure policy still admits underdiversification against state D (i.e., $g > p$). Hence, the optimal policy $\alpha_\gamma^{**}(\cdot)$ when the regulator tests for state G must have the similar two-cutoff structure to the baseline analysis. Furthermore, the regulator should unconditionally disclose its information when the regulator's model tests for state D . Specifically, releasing the information does not only perfectly identify type δ banks so that they can cheaply finance their project with a scarcity premium, but it also addresses the ex-post adverse selection for type δ banks.

Second, the probability that the regulator discloses her information – represented by the unshaded region in the left panel of Fig. 3 – increases as the average bias of the stress tests falls towards $\mu = 0.5$. That is, the model monoculture problem becomes less important in the design of optimal disclosure policy when the bias of the stress tests becomes more unpredictable. This can also be seen by the fact that g , the equilibrium fraction of type γ banks under the optimal policy,

converges to the unconstrained optimal level ($g = 0.5$), and welfare increases, as μ approaches $\frac{1}{2}$. That is, a low predictable bias against type δ banks allows the regulator to disclose tests results more often without inevitably worsening diversification. This all resonates with our conjecture in Section 4.2 that the model monoculture problem is associated with the *predictable* bias in the regulator's stress tests.

5.3. Alternative timing

In our baseline model, we assume that the regulator first commits to a disclosure policy $\alpha(b)$, then good banks choose project γ or δ , then the measure of bad banks b is realized, and finally the regulator discloses a fraction $\alpha(b)$ of good banks. In this setup, when each good bank chooses its project, it takes into account expected profits across all possible realizations of b . One might wonder whether it would have been better to assume that b is realized first, the regulator chooses a disclosure probability $\alpha(b)$ next, and finally good banks choose projects γ and δ . In this alternative setup, good banks would be able to adjust their financial projects in real time in response to both the degree of adverse selection in the capital market, and the regulator's decision about whether to disclose stress test results.

We view our baseline timing assumption as more realistic than this alternative setup, since it captures the idea that a bank's choice of portfolio cannot be adjusted rapidly in response to the announcement that a particular stress test will be conducted and the results disclosed, but can be adjusted over time in response to a systematic change in policy. For example, the SCAP was announced in February 2009, and the results released in May 2009. Many aspects of the stress test were based on banks' balance sheets as of December 31 2008.⁸ Thus, it would not have been possible for a bank to change its portfolio in order to improve its test results in response to the February 2009 announcement.

Notwithstanding these concerns, adopting the alternative timing assumption would amplify the adverse effect of disclosure on diversification. In fact, under the alternative timing it would *never* be optimal for the regulator to disclose stress test results.

Theorem 3. $\alpha^{**}(b) = 0$ for every $b \geq 0$ is weakly optimal when the fraction of bad banks is realized first, the regulator commits to a disclosure policy next, and good banks choose their projects last.

Proof. See Appendix A.4. $\square$

⁸ See <https://www.federalreserve.gov/bankinforeg/bcreg20090424a1.pdf>.

Under the alternative timing, the regulator chooses α , and thus implements g state by state to maximize $U(g, \alpha, b)$ subject to the constraint $\Delta(g, \alpha, b) = 0$. If b and $\alpha(b)$ are high enough that failing banks are rationed, the constraint $\Delta(g, \alpha, b) = 0$ implies that g must converge to 1: banks know that project δ will make zero profits, so all of them will choose γ . Such an allocation of financial projects is never optimal since $g = 1$ implies $U(g, \alpha, b) = -\infty$. Thus, the regulator will refrain from any policy that causes credit rationing to type δ banks in any degree. This in turn implies that social welfare equals $U(g, \alpha, b) = p \log g + (1 - p) \log(1 - g) + (1 + b)x$, which is maximized by setting $g = p$. The regulator can implement this allocation by never disclosing, i.e., $\alpha = 0$.

6. Related literature

A growing theoretical literature discusses the benefits and costs of disclosing stress test results or other information held by regulators. As surveyed by Goldstein and Sapra (2013) and Leitner (2014), there are a number of arguments against unconditional disclosure of stress test results: full disclosure of stress test results may reduce risk-sharing à la Hirshleifer (1971) (Goldstein and Leitner, 2018); forcing firms to disclose their financial status too often may encourage short-termism (Gigler et al., 2014); the public information provided by stress tests may crowd out private information held by individual creditors (Morris and Shin, 2002; Bond and Goldstein, 2015); and releasing the public information may cause coordination failure among banks or market participants (Bouvard et al., 2015; Inostroza and Pavan, 2017). In particular, Williams (2017) presents a model in which even though a regulator has perfect information about banks, it is optimal to only release just enough information to prevent a bank run.⁹ We complement this literature by focusing on another potential cost of stress tests, the impacts of disclosure on banks' investments and risk assessments. In particular, we study how the disclosure of stress test results may distort diversification in the entire financial system when stress tests can misclassify some financially sound banks as risky.

Another recent paper that argues that limited disclosure of public information may be optimal is Quigley and Walther (2018). They consider an environment in which market "insiders" can choose to release verifiable private information at a cost. Although the disclosure of "outside" information by the public authority can crowd out "inside" information, such crowding out can be socially desirable because disclosure of inside information can separate high-quality banks from the rest, causing inefficient bank runs. We instead focus on the way in which information disclosure affects banks' investment decisions rather than their communication with investors (which is ruled out in our setting).

Leitner and Yilmaz (2019) also consider a similar environment in which banks choose whether to privately produce information with their own risk models, given that regulators decide whether to monitor the regulated banks' information. It may be (constrained) optimal not to monitor the banks' models because monitoring would reduce the banks' incentives to collect information *ex ante*. While we abstract from the production of information either by banks or by regulators, our main result resonates with Leitner and Yilmaz (2019): full disclosure of the regulators' (imperfect) information may adversely influence banks' choices of investments, which are socially valuable but not easily recognized as such. However, an important difference is that their model is applicable to settings in which regulators rely on the banks' models (e.g. the internal ratings-based approach to calculate regulatory

capital). We instead assume regulators use their own model to regulate banks, and banks do not have the option of using their own models instead. This assumption is more applicable to the CCAR and DFAST in which the Federal Reserve conducts supervisory stress tests using its own scenarios and models (though these stress tests do use data supplied by the firms). While banks are also required to conduct their own company-run stress tests under the DFAST and CCAR, these tests cannot be used as a substitute for the supervisory stress tests. The Fed can object to banks' capital plans based on whether they fail the CCAR stress tests using the Fed: subject banks cannot use their own test results to avoid being required to take these corrective actions.

Our results suggest that regulators should keep their models secret; otherwise, banks can game the regulators' model, reducing diversification. Leitner and Williams (2023) argue that secrecy may not, in fact, be optimal because it can deter banks from making risky investments. We abstract from this possibility since banks always make some investment, whether in good or diversifying projects. However, we highlight an additional cost that arises when banks can game the regulator's model. Banks do not only invest in excessively risky projects; rather, they also invest in particular types of projects that are difficult to identify as such, reducing diversification in the overall economy.

Our framework is consistent with the role that stress tests played during the 2007–09 Global Financial Crisis. A major goal of SCAP was to alleviate widespread adverse selection problems in financial markets during the crisis (Geithner, 2015, pp. 286–287). The academic literature (e.g. Tirole (2012)) also points to adverse selection as a key reason for the lending freeze observed during the crisis. Flannery et al. (2013) document empirically that adverse selection problems worsened during the crisis.

While our result that releasing public information reduces diversification, may seem reminiscent of Morris and Shin (2002), the mechanism in our paper is different. In Morris and Shin (2002), public information reduces diversification because agents' actions are strategic complements, and they coordinate on the public signal. In our paper, banks have no intrinsic coordination motive. If anything, they have an intrinsic motive for diversification because risk-averse households reward banks for investing in assets that pay off when consumption is low. However, the disclosure of public information causes outside investors to rationally punish banks for investing differently from the herd because those banks may be risky.

Our paper relates to a growing theoretical literature that studies the causes and consequences of increased similarity in the portfolios of financial institutions. In Farhi and Tirole (2012), imperfectly targeted policy interventions make banks' leverage choices strategic complements: banks only wish to be exposed to a common shock if other institutions are also exposed, implying that policymakers will bail them out in a crisis. Kopytov (2018) presents a dynamic model in which banks' portfolios become more similar at the end of credit expansions, as these portfolios become more diversified at the individual level (see also Khorrami (2019) for a model in which an increase in diversification at the level of an individual intermediary can increase aggregate risk). We focus on a different potential cause of correlated portfolios across banks, namely the disclosure of stress test results.

Bräuning and Fillat (2019) provide the most relevant empirical evidence supporting our theoretical result that stress test disclosure can increase portfolio similarity. They find that the portfolios of U.S. banks who became subject to Dodd-Frank Act Stress Testing (DFAST) starting in 2011 became more similar since then, while the portfolios of banks not subject to DFAST remained essentially unchanged. In particular, banks with poor stress test results tend to have dissimilar portfolios before the test; after failing the test, they adjust their portfolios to become closer to the portfolios of the banks with good DFAST results. This is consistent with our theoretical finding that supervisory stress testing may create a "model monoculture" by inducing banks to choose similar business and risk models in order to avoid failing the tests, inefficiently increasing the financial system's exposure to the same aggregate risks.

⁹ The recent literature also emphasizes that the effectiveness and optimal design of stress tests depends on what other instruments are available to policymakers. Spargoli (2013) argues that disclosure of negative stress test results either reduces lending or requires costly bailouts. Similarly, Faria-e Castro et al. (2016) argue that disclosure can create inefficient bank runs unless there is a fiscal backstop.

7. Conclusion

Stress tests have moved from an exceptional measure of crisis management to a routine part of financial regulation. In a crisis, stress tests can reduce uncertainty about banks' financial soundness and prevent market failure caused by financial instability. However, routine stress tests may also lead to model monoculture, in which banks mimic the regulators' models in order to pass the tests while ignoring their own measures of risk. This can leave the financial system less diversified and vulnerable to those risks that the regulators' model ignores. We presented a simple model to understand the tradeoff facing financial regulators that use stress tests as a routine policy tool. When stress test results are predictably biased against banks holding a certain class of financial assets, our analysis suggests that the stress test results should be released to the public only selectively, either in a severe crisis or when stress test scenarios accurately evaluate the financial health of tested banks.

Our paper mainly focused on the direct impacts of releasing stress test results on portfolio adjustment decisions by subject banks. One should not hastily generalize these results to the overall effect of supervisory stress tests before carefully examining their potential indirect

effects. Disclosure of stress test results may improve financial stability through indirect channels which are not captured by our paper. In particular, the subject banks' strategic response to the stress testing can endogenously influence businesses of the other banks not subject to the supervisory stress testing. For instance, while supervisory stress testing increases the cost of lending to risky businesses for subject banks, this can create a positive selection effect where only banks with expertise in assessing the financial risks or sufficiently capitalized banks maintain or increase lending to the risky businesses (Cortés et al., 2020). Exploring such tradeoffs is beyond the scope of our paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Proofs

A.1. Proofs for Section 3

Proof of Lemma 1. As shown in Online Appendix B, each investor must symmetrically place the same buy orders in each of the submarkets labeled $\sigma = 0$ and $\sigma = 1$ in equilibrium. Let (ϕ_0^*, R_0^*) and (ϕ_1^*, R_1^*) denote the equilibrium buy orders in $\sigma = 0$ and $\sigma = 1$ submarkets, respectively. Fix a disclosure policy α and a fraction of type β banks b . Recall that at $t = 1$, each investor as a potential debt holder has the following expected utility

$$p \log(y_G + \phi_1 \alpha g R_1 x + \phi_0 (1 - \alpha) g R_0 x) + (1 - p) \log(y_D + \phi_0 (1 - g) R_0 x) - \phi_0 (1 - \alpha g + b) x - \phi_1 \alpha g x,$$

where $y_G := \phi_1 g (1 - R_1 x) + \phi_0 (1 - \alpha) g (1 - R_0 x)$ and $y_D := (1 - g) \phi_0 (1 - R_0 x)$.

Consider the submarket for the banks labeled $\sigma = 0$ and fix $\phi_1 = 1$ (we will prove $\phi_1^* = 1$ later). Then each investor, as a debt buyer, finds it optimal to offer (ϕ_0, R_0) if and only if the following first order condition with respect to ϕ_0 is satisfied:

$$p \frac{(1 - \alpha) g R_0 x}{y_G + \phi_1 \alpha g R_1 x + \phi_0 (1 - \alpha) g R_0 x} + (1 - p) \frac{(1 - g) R_0 x}{y_D + \phi_0 (1 - g) R_0 x} - (1 - \alpha g + b) x \geq 0.$$

By plugging the expressions for y_G and y_D into the above inequality, we have the following condition:

$$p \frac{(1 - \alpha) g R_0 x}{\alpha g + (1 - \alpha) g \phi_0} + (1 - p) \frac{(1 - g) R_0 x}{(1 - g) \phi_0} - (1 - \alpha g + b) x \geq 0. \quad (\text{A.1})$$

First, fix any α and b such that $(p - g)x\alpha + xb \leq 1 - x$. We first show that $(\phi_0^*, R_0^*) = \left(1, \frac{(1 - \alpha)g + (1 - g)b}{(1 - \alpha)p + (1 - p)}\right)$ is the only feasible equilibrium buy order. Note that the condition (A.1) is binding when $(\phi_0, R_0) = (\phi_0^*, R_0^*)$. Suppose to the contrary that there is an equilibrium in which the investors place a buy order $(\phi'_0, R'_0) \neq (\phi_0^*, R_0^*)$. If $\phi'_0 \leq \phi_0^*$ but $R'_0 \geq R_0^*$, we must have $p \frac{(1 - \alpha)g R'_0 x}{\alpha g + (1 - \alpha)g \phi'_0} + (1 - p) \frac{(1 - g) R'_0 x}{(1 - g) \phi'_0} - (1 - \alpha g + b) x > 0$, so an investor can profitably deviate by placing a buy order $((\phi'_0 + \epsilon) \wedge 1, R'_0 - \epsilon')$ for some sufficiently small $\epsilon, \epsilon' > 0$. If $\phi'_0 = \phi_0^*$ but $R'_0 < R_0^*$, we have $p \frac{(1 - \alpha)g R'_0 x}{\alpha g + (1 - \alpha)g \phi'_0} + (1 - p) \frac{(1 - g) R'_0 x}{(1 - g) \phi'_0} - (1 - \alpha g + b) x < 0$. Therefore, an investor can profitably deviate by offering $(\tilde{\phi}, R'_0)$, where $\tilde{\phi}$ is determined by

$$p \frac{(1 - \alpha)g R'_0 x}{\alpha g + (1 - \alpha)g \tilde{\phi}} + (1 - p) \frac{(1 - g) R'_0 x}{(1 - g) \tilde{\phi}} - (1 - \alpha g + b) x = 0.$$

If $\phi'_0 < \phi_0^* = 1$ and $R'_0 < R_0^*$, some banks with a positive measure $(1 - \phi'_0)(\alpha g + (1 - \alpha)(1 - g) + b) > 0$ will not be funded at the $\sigma = 0$ submarket. Hence, an investor can get a strictly higher payoff by placing a buy order $(\tilde{\phi}, \tilde{R}) = (1, R'_0 + \epsilon)$ for some $\epsilon > 0$ than she would by offering (ϕ'_0, R'_0) . We next show that no investor has a strict incentive to deviate from (ϕ_0^*, R_0^*) when every else investor symmetrically places the same buy order (ϕ_0^*, R_0^*) . Since all banks labeled $\sigma = 0$ is fully funded (i.e., $\phi_0^* = 1$), a deviating buy order (ϕ'_0, R'_0) is attractive to the $\sigma = 0$ banks only if $R'_0 < R_0^*$. However, since (A.1) is binding when $(\phi_0, R_0) = (\phi_0^*, R_0^*)$, we must have

$$p \frac{(1 - \alpha)g R'_0 x}{\alpha g + (1 - \alpha)g} + (1 - p) \frac{(1 - g) R'_0 x}{(1 - g)} - (1 - \alpha g + b) x < 0.$$

Therefore, to satisfy (A.1), we must have $\phi'_0 < 1$. However, we have

$$\begin{aligned}
& p \log(y_G + \alpha g R_1 x + \phi'_0(1-\alpha)gR'_0x) + (1-p) \log(y_D + \phi'_0(1-g)R'_0x) - \phi'_0(1-\alpha g+b)x - \alpha g x \\
& < p \log(y_G + \alpha g R_1 x + \phi'_0(1-\alpha)gR^*_0x) + (1-p) \log(y_D + \phi'_0(1-g)R^*_0x) - \phi'_0(1-\alpha g+b)x - \alpha g x \\
& < p \log(y_G + \alpha g R_1 x + \phi^*_0(1-\alpha)gR^*_0x) + (1-p) \log(y_D + \phi^*_0(1-g)R^*_0x) - \phi^*_0(1-\alpha g+b)x - \alpha g x,
\end{aligned}$$

where the first inequality follows from $R'_0 < R^*_0$ and the second inequality follows from the fact that the first order condition (A.1) is strictly decreasing in ϕ_0 and is binding at $\phi_0 = \phi^*_0$ conditional on $R_0 = R^*_0$. Therefore, any investor cannot profitably deviate from the prescribed equilibrium buy order (ϕ^*_0, R^*_0) .

Second, fix any α and b such that $(p-gx)\alpha + xb > 1-x$, where $R^*_0 = \frac{1}{x}$ and $\phi^*_0 < 1$ is uniquely and implicitly determined by (6) (note that (A.1) is binding when $(\phi_0, R_0) = (\phi^*_0, R^*_0)$). We first show that (ϕ^*_0, R^*_0) is the only feasible equilibrium buy order. Suppose to the contrary that there exists an equilibrium in which the investors place the buy order $(\phi', R') \neq (\phi^*_0, R^*_0)$. If $\phi'_0 \geq \phi^*_0$ but $R'_0 < R^*_0$, then we have $p \frac{(1-\alpha)gR'_0x}{\alpha g + (1-\alpha)g\phi'_0} + (1-p) \frac{(1-g)R'_0x}{(1-g)\phi'_0} - (1-\alpha g+b)x < 0$, so an investor can profitably deviate by offering $(\tilde{\phi}_0, \tilde{R}_0)$, where $\tilde{R} = R'_0$ and $\tilde{\phi}_0$ is determined by

$$p \frac{(1-\alpha)gR'_0x}{\alpha g + (1-\alpha)g\tilde{\phi}_0} + (1-p) \frac{(1-g)R'_0x}{(1-g)\tilde{\phi}_0} - (1-\alpha g+b)x = 0.$$

If $\phi'_0 < \phi^*_0$ and $R'_0 = R^*_0$, then we have $p \frac{(1-\alpha)gR'_0x}{\alpha g + (1-\alpha)g\phi'_0} + (1-p) \frac{(1-g)R'_0x}{(1-g)\phi'_0} - (1-\alpha g+b)x > 0$, and thus an investor can profitably deviate by placing a buy order $(\tilde{\phi}_0, \tilde{R}_0) = (\phi'_0 + \varepsilon, R'_0)$ for some $\varepsilon > 0$. If $\phi'_0 < \phi^*_0$ and $R'_0 < R^*_0$, some banks with a positive measure $(\phi^*_0 - \phi'_0)(\alpha g + (1-\alpha)(1-g) + b) > 0$ will not be funded at the $\sigma = 0$ submarket. Hence, an investor can get a strictly higher payoff by placing a buy order $(\tilde{\phi}_0, \tilde{R}_0) = (\phi^*_0, R'_0 + \varepsilon)$ for some $\varepsilon > 0$ than she would by offering (ϕ'_0, R'_0) .

We next show that no investor has a strict incentive to deviate from (ϕ^*_0, R^*_0) when every else investor symmetrically places the same buy order (ϕ^*_0, R^*_0) . Since $R^*_0 = \frac{1}{x}$, any deviating offer (ϕ'_0, R'_0) must satisfy $R'_0 < R^*_0$, which implies $\phi'_0 < \phi^*_0$ to satisfy (A.1). However, by applying the similar logic used in the previous case $(p-gx)\alpha + xb \leq 1-x$, one can show that the deviating investor will always get a strictly lower utility than she would by placing the order (ϕ^*_0, R^*_0) .

We last show that $\frac{d\phi^*_0}{d\alpha} < 0$ and $\frac{d\phi^*_0}{db} < 0$ for every α and b such that $(p-gx)\alpha + xb > 1-x$. By applying the Implicit Function Theorem with respect to α , we have

$$\frac{d\phi^*_0}{d\alpha} = - \frac{p \left(\frac{1}{\alpha + (1-\alpha)\phi^*_0} \right)^2 - gx}{p \frac{1}{\left(\frac{\alpha}{1-\alpha} + \phi^*_0 \right)^2} + (1-p) \frac{1}{(\phi^*_0)^2}} < 0,$$

where the inequality holds true because the numerator of the derivative function is positive, which follows from $p-x > 0$, $g \in (0, 1)$, and $\frac{1}{\alpha + (1-\alpha)\phi^*_0} > 1$. Similarly, by applying the Implicit Function Theorem with respect to b , we have

$$\frac{d\phi^*_0}{db} = - \frac{x}{p \frac{1}{\left(\frac{\alpha}{1-\alpha} + \phi^*_0 \right)^2} + (1-p) \frac{1}{(\phi^*_0)^2}} < 0.$$

Next, consider the submarket for type γ banks labeled $\sigma = 1$ by the regulator. Similar to the $\sigma = 0$ submarket, given the equilibrium buy order (ϕ^*_0, R^*_0) , each investor finds it optimal to place a buy order (ϕ_1, R_1) to the $\sigma = 1$ banks only if the following first order condition with respect to ϕ_1 satisfies:

$$p \frac{\alpha g R_1 x}{\alpha g \phi_1 + (1-\alpha)g\phi^*_0} - \alpha g x \geq 0. \quad (\text{A.2})$$

Let $(\phi^*_1, R^*_1) = \left(1, \frac{(\alpha + (1-\alpha)\phi^*_0)g}{p} \right)$. Note that the condition (A.2) is binding when $(\phi_1, R_1) = (\phi^*_1, R^*_1)$. Then, by applying the similar logic used to establish the equilibrium in the $\sigma = 0$ submarket as above, one can show that (ϕ^*_1, R^*_1) is the unique equilibrium buy order at the $\sigma = 1$ submarket. $\square$

Proof of Corollary 1. The economy with zero disclosure is indeed a special case of $\alpha = 0$ where every bank is labeled as $\sigma = 0$. Thus, the date-1 equilibrium has the same outcome as the $\sigma = 0$ submarket with $\alpha = 0$, where $(\phi^*, R^*) = (\phi^*_0, R^*_0)|_{\alpha=0}$ for each $b \geq 0$. $\square$

Proof of Lemma 2. To prove part (i), we first derive $\frac{d}{db}[U(g, \alpha, b) - U(g, 0, b)]$. By applying the Implicit Function Theorem, we have

$$\frac{d}{db}[U(g, \alpha, b) - U(g, 0, b)] = x(\phi^* - \phi^*_0). \quad (\text{A.3})$$

Since $\phi^* = \phi^*_0 = 1$ if $b \leq \left(\frac{1}{x} - 1 \right) - \alpha \left(\frac{p}{x} - g \right)$, $1 = \phi^* > \phi^*_0$ if $\left(\frac{1}{x} - 1 \right) - \alpha \left(\frac{p}{x} - g \right) < b \leq \frac{1}{x} - 1$, and $1 > \phi^* > \phi^*_0$ if $b > \frac{1}{x} - 1$, we have $\frac{d}{db}[U(g, \alpha, b) - U(g, 0, b)] \geq 0$ for every $b \geq 0$, where the inequality strictly holds for every $b > \left(\frac{1}{x} - 1 \right) - \alpha \left(\frac{p}{x} - g \right)$.

To prove part (ii), we first derive $\frac{d}{d\alpha}[U(g, \alpha, b) - U(g, 0, b)]$. By applying the Implicit Function Theorem, we have

$$\frac{d}{d\alpha}[U(g, \alpha, b) - U(g, 0, b)] = (1 - \phi^*_0) \left[\frac{1}{\alpha + (1-\alpha)\phi^*_0} p - gx \right].$$

Since $\frac{1}{\alpha+(1-\alpha)\phi_0^*}p - gx > p - gx > p - x > 0$, the sign of the derivative depends on the first parenthesized term $1 - \phi_0^*$. Since $\phi_0^* = 1$ if $\alpha \leq \left[0 \vee \left(\frac{1-x(1+b)}{p-gx}\right)\right] \wedge 1$ and $\phi_0^* < 1$ otherwise, we have $\frac{d}{d\alpha} [U(g, \alpha, b) - U(g, 0, b)] \geq 0$, where the inequality strictly holds for $\alpha > \left[0 \vee \left(\frac{1-x(1+b)}{p-gx}\right)\right] \wedge 1$. $\square$

A.2. Proofs for Section 4

We first introduce a new threshold value $\hat{\alpha}$, which will be frequently used in the proofs. Formally, for a given $b > \frac{1-p}{x} - (1-g)$, define $\hat{\alpha} \in [0, 1]$ as

$$\hat{\alpha} := 0 \vee \left[\left(1 - \frac{\frac{1-p}{x} - [(1-g) + b]}{\left(\frac{p}{x} - g\right)} \right) \wedge 1 \right]. \quad (\text{A.4})$$

Then, $\phi_0^* = 1$ if $\alpha \leq \hat{\alpha}$ and $\phi_0^* < 1$ otherwise. For every $\alpha \leq \hat{\alpha}$, we have $U(g, \alpha, b) = U(g, 0, b)$, whereas $U(g, \alpha, b) > U(g, 0, b)$ otherwise. Furthermore, the following mathematical properties are immediate by construction.

Lemma A.1.

- (i) $\hat{\alpha} = 1$ at $b = \frac{1-p}{x} - (1-g)$ and $\hat{\alpha} = 0$ at $b = \frac{1}{x} - 1$;
- (ii) $\hat{\alpha}$ is strictly decreasing in b for all $b \in \left[\frac{1-p}{x} - (1-g), \frac{1}{x} - 1\right]$.

Next, we derive an explicit mathematical formula of $\Delta(g, \alpha, b)$. As was seen in Section 4, the $t = 0$ utility function of each good bank is

$$p \log(\alpha g(1 - R_1^*x) + (1-\alpha)g\phi_0^*(1 - R_0^*x) + z_G) + (1-p) \log((1-g)\phi_0^*(1 - R_0^*x) + z_D) - C,$$

The marginal utility of choosing type γ rather than δ equals the first order derivative with respect to g , taking z_G , z_D , and C as given:

$$p \frac{\alpha(1 - R_1^*x) + (1-\alpha)\phi_0^*(1 - R_0^*x)}{\alpha g(1 - R_1^*x) + (1-\alpha)g\phi_0^*(1 - R_0^*x) + z_G} - (1-p) \frac{\phi_0^*(1 - R_0^*x)}{(1-g)\phi_0^*(1 - R_0^*x) + z_D}.$$

By plugging in $z_G = \alpha g R_1^*x + (1-\alpha)g\phi_0^*R_0^*x$ and $z_D = (1-g)\phi_0^*R_0^*x$, we can derive $\Delta(g, \alpha, b)$ as follows:

$$\Delta(g, \alpha, b) := \alpha \frac{p}{g} \left(\frac{1}{\alpha + (1-\alpha)\phi_0^*} - \frac{g}{p}x \right) + \left[\left(\frac{p}{g} \right) \frac{(1-\alpha)\phi_0^*}{\alpha + (1-\alpha)\phi_0^*} - \frac{1-p}{1-g} \right] (1 - R_0^*x). \quad (\text{A.5})$$

Proof of Lemma 3. We first show that $g \geq p$ under any feasible policy. Suppose to the contrary that some optimal disclosure policy induces $g < p$. Recall that $\Delta(g, \alpha, b)$ has different forms, depending on whether $\phi_0^* = 1$ or $\phi_0^* < 1$. Consider $\phi_0^* = 1$ first. Recall that $\phi_0^* = 1$ if either $b \leq \frac{1-p}{x} - (1-g)$ or $b \in \left(\frac{1-p}{x} - (1-g), \frac{1}{x} - 1\right]$ and $\alpha \leq \hat{\alpha}$. By plugging $(\phi_1^*, R_1^*) = \left(1, \frac{(1-\alpha)g}{p}\right)$ and $(\phi_0^*, R_0^*) = \left(1, \frac{(1-\alpha)g + (1-g)b}{(1-\alpha)p + (1-p)}\right)$ into (A.5), we have

$$\Delta(g, \alpha, b) = \left[\left(\frac{p}{g} - \frac{1-p}{1-g} \right) - \left(1 + \frac{(1-g)+b}{g} \right)x \right] + \frac{x(1-p)}{p(1-g)} \left(1 + \frac{\frac{(1-g)+b}{g} - \frac{1-p}{p}}{(1-\alpha) + \frac{1-p}{p}} \right).$$

$\Delta(g, \alpha, b)$ expressed as above is increasing in α if and only if $\frac{(1-g)+b}{g} - \frac{1-p}{p} > 0$, or equivalently, $b > \frac{g}{p} - 1$. Since $g < p$, we have $\frac{g}{p} < 1$. Hence, $\Delta(g, \alpha, b)$ is increasing in α for every $b \leq \frac{1}{x} - 1$ and $\alpha \leq \hat{\alpha}$. Furthermore, $g < p$ implies $\Delta(g, \alpha, 0) > 0$ for every $\alpha \in [0, 1]$. Lastly, $\Delta(g, 0, b) = \left(\frac{p}{g} - \frac{1-p}{1-g}\right)[1 - (1+b)x] \geq 0$ for every $b \leq \frac{1}{x} - 1$, where the equality holds only for the case $b = \frac{1}{x} - 1$. From all the observations above, we have $\Delta(g, \alpha, b) \geq 0$ for every (α, b) that yields $\phi_0^* = 1$, where the equality holds only for the case $b = \frac{1}{x} - 1$ and $\alpha = 0$.

Consider the case $\phi_0^* < 1$ next. Since $R_0^* = \frac{1}{x}$, we have $\Delta(g, \alpha, b) = \alpha \left[\frac{1}{\alpha + (1-\alpha)\phi_0^*} \left(\frac{p}{g} \right) - x \right]$. Since $p - x > 0$, we have $\Delta(g, \alpha, b) > 0$ for every $\alpha > 0$ and $\Delta(g, 0, b) = 0$ at $\alpha = 0$. Combining all the observations together, we have $\int_0^\infty \Delta(g, \alpha(b), b) dF(b) > 0$ for any disclosure policy $\alpha(b)$, which contradicts (9).

Next, suppose $g = p$. By applying the same logic used in the previous case $g < p$, one can find that $\Delta(g, \alpha, b) > 0$ for every $b \geq 0$ if $\alpha > 0$, whereas $\Delta(g, 0, b) = 0$ for every $b \geq 0$. Hence, to support $g = p$, we must have $\alpha(b) = 0$ with probability 1.

Finally, suppose by contradiction that $g = p$ under an optimal disclosure policy, implying that $\alpha(b) = 0$ with probability 1. Consider a deviation in which $\alpha(b) = 1$ for $b \in \left(\frac{1}{x} - 1, \frac{1}{x} - 1 + \varepsilon\right)$ for $\varepsilon > 0$ sufficiently small. For given ε , $g(\varepsilon)$ is determined as a differentiable function of ε by (9), which becomes

$$\int_0^\infty \Delta(g(\varepsilon), 0, b) dF(b) + \int_{\frac{1}{x}-1}^{\frac{1}{x}-1+\varepsilon} [\Delta(g(\varepsilon), 1, b) - \Delta(g(\varepsilon), 0, b)] dF(b) = 0$$

Welfare under this deviation is

$$\int_0^{\frac{1}{x}-1+\varepsilon} U(g(\varepsilon), 0, b) dF(b) + \int_{\frac{1}{x}-1}^{\frac{1}{x}-1+\varepsilon} [U(g(\varepsilon), 1, b) - U(g(\varepsilon), 0, b)] dF(b) = 0$$

The derivative of welfare with respect to ε , evaluated at $\varepsilon = 0$, equals

$$\left[U\left(p, 1, \frac{1}{x} - 1\right) - U\left(p, 0, \frac{1}{x} - 1\right) \right] f\left(\frac{1}{x} - 1\right) > 0$$

This implies that the deviation increases welfare for small enough $\varepsilon > 0$, which contradicts the original allocation being optimal. $\square$

Proof of Lemma 4. We first derive some useful mathematical properties of Δ . First, we show $\frac{d\Delta}{dg} < 0$. If $\phi_0^* = 1$, $\Delta(g, \alpha, b)$ is rewritten as

$$\Delta(g, \alpha, b) = \left(\frac{p}{g} - \frac{1-p}{1-g} \right) (1 - R_0^* x) + \alpha x (R_0^* - 1).$$

Moreover, we have $\frac{dR_0^*}{dg} < 0$ since $R_0^* = \frac{(1-\alpha)g + (1-g)b}{(1-\alpha)p + (1-p)}$. Furthermore, we have $g > p$ by Lemma A.1. In sum, we have $\frac{d\Delta}{dg} < 0$. If $\phi_0^* < 1$, we have

$$\Delta(g, \alpha, b) = \alpha \left(\frac{p}{g} \left\{ \frac{1}{\alpha + (1-\alpha)\phi_0^*} \right\} - x \right).$$

By applying the implicit function theorem to (6), we have $\frac{d\phi_0^*}{dg} > 0$ whenever $\phi_0^* < 1$. Hence we have $\frac{d\Delta}{dg} < 0$. Second, we show $\frac{dU}{dg} \leq 0$. If $\phi_0^* = 1$, we have $U(g, \alpha, b) = -(1+b)x$, and therefore, $\frac{dU}{dg} = 0$. If $\phi_0^* < 1$, we have $\frac{dU}{dg} = \frac{\partial U}{\partial g} = -\alpha(1 - \phi_0^*)x < 0$, where the first equality follows from the envelope theorem.

Throughout, fix an optimal disclosure policy $\alpha(\cdot)$ such that $\mathbb{E}[\alpha(b)] > 0$. By Lemma 3, we have $g > p$, which implies $\frac{g}{p} - 1 > 0$. Furthermore, by Lemma A.1, we have

$$\Delta(g, \alpha, b) = \left[\left(\frac{p}{g} - \frac{1-p}{1-g} \right) - \left(1 + \frac{(1-g)+b}{g} \right) x \right] + \frac{x(1-p)}{p(1-g)} \left(1 + \frac{\frac{(1-g)+b}{g} - \frac{1-p}{p}}{(1-\alpha) + \frac{1-p}{p}} \right)$$

for each $b \leq \frac{1-p}{x} - (1-g)$. As was seen in the proof of Lemma 3, $\Delta(g, \alpha, b)$ is decreasing in α if $b \leq \frac{g}{p} - 1$ and increasing in α if $b \in \left(\frac{g}{p} - 1, \frac{1-p}{x} - (1-g) \right]$. Lastly, $U(g, \alpha, b) - U(g, 0, b) = 0$ for all $b \leq \frac{1-p}{x} - (1-g)$ and $\alpha \in (0, 1]$.

To show optimality of $\alpha(b) = 1$ for all $b \leq \frac{g}{p} - 1$, suppose to the contrary that $\alpha(b) < 1$ for a positive measure of $b \leq \frac{g}{p} - 1$. Since $\frac{d\Delta}{dg} < 0$ and $\frac{dU}{dg} \leq 0$, the regulator can increase welfare by making g closer to p by switching the disclosure policy to a $\alpha'(b) = (\alpha(b) + \varepsilon) \wedge 1$ for a small $\varepsilon > 0$ and every $b \leq \frac{g}{p} - 1$. Such a policy can decrease g since $\Delta(g, \alpha, b)$ is decreasing in α for every $b \leq \frac{g}{p} - 1$, whereas the change in α does not directly affect welfare since $U(g, \alpha', b) = U(g, \alpha, b)$ for every $b \leq \frac{g}{p} - 1$.

Next, to show optimality of $\alpha(b) = 0$ for all $b \in \left(\frac{g}{p} - 1, \frac{1-p}{x} - (1-g) \right]$, suppose to the contrary that $\alpha(b) > 0$ for a positive measure of $b \in \left(\frac{g}{p} - 1, \frac{1-p}{x} - (1-g) \right]$. Then, by changing the disclosure policy to $\alpha'(b) = (\alpha(b) - \varepsilon) \vee 0$ for a small $\varepsilon > 0$ and every $b \in \left(\frac{g}{p} - 1, \frac{1-p}{x} - (1-g) \right]$. Since Δ is increasing in α for every $b \in \left(\frac{g}{p} - 1, \frac{1-p}{x} - (1-g) \right]$ and Δ is increasing in g , such a policy can indirectly increase welfare by bringing g closer to p , whereas it does not directly change welfare since $U(g, \alpha', b) = U(g, \alpha, b)$ for every $b \in \left(\frac{g}{p} - 1, \frac{1-p}{x} - (1-g) \right]$. Putting all these observations together, any optimal disclosure policy must satisfy $\alpha(b) = 1$ if $b \leq \frac{g}{p} - 1$ and $\alpha(b) = 0$ if $b \in \left(\frac{g}{p} - 1, \frac{1-p}{x} - (1-g) \right]$. $\square$

In what follows, we present some properties of $\alpha^{**}(\cdot)$ and $W(g, \alpha, b)$, which will be used in the proof of Lemma 5.

Lemma A.2. $\alpha^{**}(b) \in \{0, 1\}$ for $b > \frac{1-p}{x} - (1-g)$.

Proof. We first prove that any $\alpha \leq \hat{\alpha}$ cannot be optimal for every $\frac{1-p}{x} - (1-g) < b < \frac{1}{x} - 1$. For any $b \in \left(\frac{1-p}{x} - (1-g), \frac{1}{x} - 1 \right]$, we have $U(g, \alpha, b) - U(g, 0, b) = 0$ but $\Delta(g, \alpha, b) > 0$, and thus $W(g, \alpha, b) < 0$ for every $\alpha \leq \hat{\alpha}$. Hence, the regulator never chooses $\alpha \leq \hat{\alpha}$. Hence, we focus on the disclosure policies $\alpha \in (\hat{\alpha}, 1]$ throughout in the proof.

Next, we introduce some new notations used throughout the proof. For $b > \frac{1-p}{x} - (1-g)$, we define $\bar{\phi}_0^* := \sup_{\alpha \geq \hat{\alpha}} \phi_0^*$ and $\underline{\phi}_0^* := \inf_{\alpha \geq \hat{\alpha}} \phi_0^*$. Since $\frac{d\phi_0^*}{d\alpha} \leq 0$, we have $\bar{\phi}_0^* = \lim_{\alpha \rightarrow \hat{\alpha}} \phi_0^* (\leq 1)$ and $\underline{\phi}_0^* = \lim_{\alpha \rightarrow 1} \phi_0^* = \frac{((1-g)+b)x}{(1-p)}$. Moreover, since $\Delta(g, \alpha, b) = \alpha \left(\frac{p}{g} \left\{ \frac{1}{\alpha + (1-\alpha)\phi_0^*} \right\} - x \right)$, $W(g, \alpha, b)$ is expressed for every $\alpha \geq \hat{\alpha}$ and $b > \frac{1-p}{x} - (1-g)$ as follows:

$$\begin{aligned} W(g, \alpha, b) &= [p \log(\alpha + (1-\alpha)\phi_0^*) - gx(\alpha + (1-\alpha)\phi_0^*)] + [(1-p) \log \phi_0^* - ((1-g)+b)x\phi_0^*] \\ &\quad - \lambda \alpha \left(\frac{1}{\alpha + (1-\alpha)\phi_0^*} \frac{p}{g} - x \right) - [p \log \bar{\phi}_0^* + (1-p) \log \bar{\phi}_0^* - (1+b)x\bar{\phi}_0^*] \\ &= \left[p \log(\alpha + (1-\alpha)\phi_0^*) - gx(\alpha + (1-\alpha)\phi_0^*) - \lambda \alpha \left(\frac{1}{\alpha + (1-\alpha)\phi_0^*} \frac{p}{g} - x \right) \right] \end{aligned}$$

$$+ [(1-p)\log \phi_0^* - ((1-g)+b)x\phi_0^*] - [p\log \bar{\phi}_0^* + (1-p)\log \bar{\phi}_0^* - (1+b)x\bar{\phi}_0^*].$$

In the last expression, the first bracketed term is increasing in ϕ_0^* since $p > gx$, and thus, maximized at $\phi_0^* = \bar{\phi}_0^*$. Furthermore, the second bracketed term is concave in ϕ_0^* , and maximized at the unique extreme point $\phi_0^* = \underline{\phi}_0^*$.

Using these properties, define $\bar{W}(g, \alpha, b)$ as follows:

$$\begin{aligned} \bar{W}(g, \alpha, b) = & \left[p\log(\alpha + (1-\alpha)\bar{\phi}_0^*) - gx(\alpha + (1-\alpha)\bar{\phi}_0^*) - \lambda\alpha \left(\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \frac{p}{g} - x \right) \right] \\ & + [(1-p)\log \underline{\phi}_0^* - ((1-g)+b)x\underline{\phi}_0^*] - [p\log \bar{\phi}_0^* + (1-p)\log \bar{\phi}_0^* - (1+b)x\bar{\phi}_0^*]. \end{aligned}$$

By construction, $\bar{W}(g, \alpha, b) \geq W(g, \alpha, b)$ for every $\alpha \in (\hat{\alpha}, 1]$, where the inequality is strict for all $\alpha \geq \hat{\alpha}$ but $\alpha = 1$. Lastly, for every $\phi_\gamma, \phi_\delta \in [\underline{\phi}_0^*, \bar{\phi}_0^*]$, define

$$\begin{aligned} \bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b) := & \left[p\log(\alpha + (1-\alpha)\phi_\gamma) - (\alpha + (1-\alpha)\phi_\gamma)gx - \frac{\lambda}{g}\alpha \left(p\frac{1}{\alpha + (1-\alpha)\phi_\gamma} - gx \right) \right] \\ & + [(1-p)\log \phi_\delta - \phi_\delta((1-g)+b)x] - [p\log \bar{\phi}_0^* + (1-p)\log \bar{\phi}_0^* - (1+b)x\bar{\phi}_0^*]. \end{aligned}$$

$\bar{W}$ is increasing in ϕ_γ and decreasing in ϕ_δ for any $\phi_\gamma, \phi_\delta \in [\underline{\phi}_0^*, \bar{\phi}_0^*]$, respectively.

Now, the proof proceeds in several claims.

Claim A.1. Both $\bar{W}(g, \alpha, b)$ and $\bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b)$ are quasi-convex in α for all $\alpha \geq \hat{\alpha}$.

Proof. We prove quasi-convexity of $\bar{W}$ only, because the proof of quasi-convexity of $\bar{W}$ is similar. Differentiating $\bar{W}$ with respect to α , we have

$$\begin{aligned} \frac{d\bar{W}}{d\alpha} &= (1-\bar{\phi}_0^*) \left(p\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} - gx \right) - \frac{\lambda}{g} \left(p\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} - gx \right) + \lambda\frac{p}{g} \frac{\alpha(1-\bar{\phi}_0^*)}{(\alpha + (1-\alpha)\bar{\phi}_0^*)^2} \\ &= (1-\bar{\phi}_0^*) \left(p\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} - gx \right) + \lambda\frac{p}{g} \left(\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \right) \left(\frac{\alpha(1-\bar{\phi}_0^*)}{\alpha + (1-\alpha)\bar{\phi}_0^*} - 1 \right) + \lambda x \\ &= (1-\bar{\phi}_0^*) \left(p\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} - gx \right) + \lambda\frac{p}{g} \left(\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \right) \left(-\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} \right) + \lambda x \\ &= \frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \cdot \frac{p}{g} \left[(1-\bar{\phi}_0^*)g - \lambda\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} \right] - ((1-\bar{\phi}_0^*)g - \lambda)x. \end{aligned}$$

Since $\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} \leq 1$, $\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \geq 1$, and $p - gx > 0$, we have

$$\left(\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \right) \frac{p}{g} > x$$

and

$$(1-\bar{\phi}_0^*)g - \lambda\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} \geq ((1-\bar{\phi}_0^*)g - \lambda).$$

If $(1-\bar{\phi}_0^*)g - \lambda\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} \geq 0$, then we always have $\frac{d\bar{W}}{d\alpha} > 0$. Consider $(1-\bar{\phi}_0^*)g - \lambda\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} < 0$ next. If $\frac{d\bar{W}}{d\alpha} = 0$, we have

$$\frac{d\bar{W}}{d\alpha} = \frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \cdot \frac{p}{g} \left[(1-\bar{\phi}_0^*)g - \lambda\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} \right] - ((1-\bar{\phi}_0^*)g - \lambda)x = 0.$$

Since $(1-\bar{\phi}_0^*)g - \lambda\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} < 0$, we have

$$\begin{aligned} \frac{d^2\bar{W}}{d\alpha^2} &= \frac{d}{d\alpha} \left[\frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \cdot \frac{p}{g} \left((1-\bar{\phi}_0^*)g - \lambda\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} \right) \right] \\ &= \frac{p}{g} \left[-\frac{(1-\bar{\phi}_0^*)}{(\alpha + (1-\alpha)\bar{\phi}_0^*)^2} \left((1-\bar{\phi}_0^*)g - \lambda\frac{\bar{\phi}_0^*}{\alpha + (1-\alpha)\bar{\phi}_0^*} \right) + \frac{1}{\alpha + (1-\alpha)\bar{\phi}_0^*} \left(\frac{\lambda\bar{\phi}_0^*(1-\bar{\phi}_0^*)}{(\alpha + (1-\alpha)\bar{\phi}_0^*)^2} \right) \right] \\ &> 0 \end{aligned}$$

if $\frac{d\bar{W}}{d\alpha} = 0$, which implies the desired result. $\square$

Claim A.2. If $\bar{W}(g, \hat{\alpha}, b) > \bar{W}(g, 1, b)$, there exists a $\bar{\alpha} \in (\hat{\alpha}, 1]$ such that $\bar{W}(g, \alpha, b) > \bar{W}(g, 1, b) \vee 0$ if and only if $\alpha \in (\hat{\alpha}, \bar{\alpha})$.

Proof. The proof is straightforward from Claim A.1. $\square$

Claim A.3. $\bar{W}(\phi, \phi, g, \hat{\alpha}, b) < W(g, \hat{\alpha}, b) (\leq 0)$ for all $\phi < \bar{\phi}_0^*$.

Proof. Since $\bar{W}(\phi, \phi, g, \hat{\alpha}, b)$ is a linear sum of two strictly concave functions of ϕ , $\bar{W}(\phi, \phi, g, \hat{\alpha}, b)$ is a globally concave function of ϕ . Furthermore, $\bar{W}(\phi, \phi, g, \hat{\alpha}, b)$ is uniquely maximized at $\phi = \bar{\phi}_0^*$, which implies $\bar{W}(\phi, \phi, g, \hat{\alpha}, b) < W(\bar{\phi}_0^*, \bar{\phi}_0^*, g, \hat{\alpha}, b) = W(g, \hat{\alpha}, b) \leq 0$. $\square$

Claim A.4. Suppose $\bar{W}(\phi_\gamma, \phi_\delta, g, \hat{\alpha}, b) = W(g, 1, b) \vee 0$ for some $\phi_\gamma \in [\underline{\phi}_0^*, \bar{\phi}_0^*]$ and $\phi_\delta \geq \underline{\phi}_0^*$. Then, $\bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b) < W(g, 1, b) \vee 0$ for all $\alpha \in [\hat{\alpha}, \bar{\alpha}]$.

Proof. Recall that $\bar{W}(g, 1, b) = W(g, 1, b)$. Suppose to the contrary that there exists a $\tilde{\alpha} \in [\hat{\alpha}, \bar{\alpha}]$ such that $\bar{W}(\phi_\gamma, \phi_\delta, g, \tilde{\alpha}, b) \geq W(g, 1, b) \vee 0 = \bar{W}(\phi_\gamma, \phi_\delta, g, \hat{\alpha}, b)$. Since $\bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b)$ is quasi-convex in α and $\bar{W}(\phi_\gamma, \phi_\delta, g, \hat{\alpha}, b) = W(g, 1, b) \vee 0$ by assumption, $\bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b)$ must be monotone increasing in α for all $\alpha \geq \tilde{\alpha}$.

There are two possible cases, $\phi_\delta > \underline{\phi}_0^*$ or $\phi_\delta = \underline{\phi}_0^*$. If $\phi_\delta > \underline{\phi}_0^*$, we must have $\bar{W}(\phi_\gamma, \phi_\delta, g, 1, b) < W(g, 1, b) \vee 0$ since $\phi_\gamma < \bar{\phi}_0^*$. Furthermore, we have $\bar{W}(\phi_\gamma, \phi_\delta, g, \tilde{\alpha}, b) \geq W(g, 1, b) > \bar{W}(\phi_\gamma, \phi_\delta, g, 1, b)$. Since $\tilde{\alpha} < 1$, $\bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b)$ must decrease in α for some $\alpha \geq \tilde{\alpha}$, which contradicts quasi-convexity of $\bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b)$.

If $\phi_\delta = \underline{\phi}_0^*$, we must have

$$\bar{W}(\phi_\gamma, \phi_\delta, g, \bar{\alpha}, b) \geq \bar{W}(\phi_\gamma, \phi_\delta, g, \tilde{\alpha}, b) \geq W(g, 1, b) \vee 0 = \bar{W}(g, \bar{\alpha}, b),$$

where the first inequality follows from the observation that $\bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b)$ is monotone increasing in α for all $\alpha \geq \tilde{\alpha}$, and the equality follows from the definition of $\bar{\alpha}$. However, since $\phi_\gamma < \bar{\phi}_0^*$, we must have

$$\bar{W}(\phi_\gamma, \phi_\delta, g, \bar{\alpha}, b) < \bar{W}(\bar{\phi}_0^*, \phi_\delta, g, \bar{\alpha}, b) = \bar{W}(g, \bar{\alpha}, b),$$

where the strict inequality follows from $\frac{\partial \bar{W}}{\partial \phi_\gamma} > 0$, and the equality follows from the definition of $\bar{W}(g, \bar{\alpha}, b)$. Combining the observations above together, we have $\bar{W}(\phi_\gamma, \phi_\delta, g, \bar{\alpha}, b) \geq \bar{W}(g, \bar{\alpha}, b) > \bar{W}(\phi_\gamma, \phi_\delta, g, \bar{\alpha}, b)$, a contradiction. $\square$

Claim A.5. Suppose $\bar{W}(g, \hat{\alpha}, b) > \bar{W}(g, 1, b) \vee 0$. Then $W(g, \alpha, b) < W(g, 1, b)$ for all $\alpha \in [0, 1)$.

Proof. Recall that $\bar{W}(g, 1, b) = W(g, 1, b)$ and $W(g, \hat{\alpha}, b) = \bar{W}(\bar{\phi}_0^*, \bar{\phi}_0^*, g, \hat{\alpha}, b) \leq 0$. By Claim A.2 and quasi-convexity of $\bar{W}$, we know

$$W(g, \alpha, b) < \bar{W}(g, \alpha, b) \leq \bar{W}(g, 1, b) \vee 0 = W(g, 1, b) \vee 0$$

for all $\alpha \in (\bar{\alpha}, 1]$. Hence, $W(g, \alpha, b) < W(g, 1, b) \vee 0$ for all $\alpha \in (\hat{\alpha}, 1)$. Furthermore, we showed above that $W(g, \alpha, b) < 0$ for every $\alpha \leq \hat{\alpha}$.

It remains to prove that $W(g, \alpha, b) < \bar{W}(g, 1, b) \vee 0$ for all $\alpha \in (\hat{\alpha}, \bar{\alpha}]$. Suppose $\bar{W}(g, \hat{\alpha}, b) > \bar{W}(g, 1, b) \vee 0$. We first prove for the case $\bar{W}(g, 1, b) > 0$. Since $\bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b)$ is decreasing in ϕ_δ and $\phi_0^*(\alpha)$ is decreasing in α , there exists a $\alpha_1 \in (\hat{\alpha}, 1)$ (and thus $\underline{\phi}_0^* < \phi_0^*(\alpha_1) < \bar{\phi}_0^*$) such that $\bar{W}(\bar{\phi}_0^*, \phi_0^*(\alpha_1), g, \hat{\alpha}, b) = W(g, 1, b)$.

We first show $\bar{W}(\bar{\phi}_0^*, \phi_0^*(\alpha_1), g, \alpha, b) \leq W(g, 1, b)$ for every $\alpha \leq \bar{\alpha} \vee \alpha_1$. Suppose to the contrary that $\bar{W}(\bar{\phi}_0^*, \phi_0^*(\alpha_1), g, \alpha', b) > W(g, 1, b)$ for some $\alpha' \leq \bar{\alpha} \vee \alpha_1$. By quasi-convexity of $\bar{W}$, we must have $\bar{W}(\bar{\phi}_0^*, \phi_0^*(\alpha_1), g, \alpha, b) > W(g, 1, b)$ for all $\alpha > \alpha'$. Since $\bar{W}(g, \alpha, b)$ is decreasing in α for all $\alpha \leq \bar{\alpha}$ by definition of $\bar{\alpha}$, we must have $\bar{W}(\bar{\phi}_0^*, \phi_0^*(\alpha_1), g, \alpha, b) > \bar{W}(g, \alpha, b)$ for some $\alpha \in \alpha'$, which contradicts $\bar{W} \leq \bar{W}$ for every α . From this observation, we have

$$W(g, \alpha, b) = \bar{W}(\phi_0^*(\alpha), \phi_0^*(\alpha), g, \alpha, b) < \bar{W}(\bar{\phi}_0^*, \phi_0^*(\alpha_1), g, \alpha, b) \leq W(g, 1, b) \quad (\text{A.6})$$

for all $\alpha \leq \alpha_1$, where the strict inequality follows from $\phi_0^*(\alpha_1) < \phi_0^*(\alpha) \leq \bar{\phi}_0^*$ for all $\hat{\alpha} \leq \alpha < \alpha_1$, $\frac{\partial \bar{W}}{\partial \phi_\gamma} > 0$, and $\frac{\partial \bar{W}}{\partial \phi_\delta} < 0$. If $\alpha_1 \geq \bar{\alpha}$, the proof is done since $W(g, \alpha, b) < W(g, 1, b)$ for all $\alpha \leq \bar{\alpha}$ by (A.6). Hence we restrict our attention to $\alpha_1 < \bar{\alpha}$.

Since $\bar{W}(\phi_0^*(\alpha_1), \phi_\delta, g, \hat{\alpha}, b)$ is decreasing in ϕ_δ for all $\phi_\delta \in [\underline{\phi}_0^*, \bar{\phi}_0^*]$, there are two possible cases: (i) $\bar{W}(\phi_0^*(\alpha_1), \phi_0^*(\bar{\alpha}), g, \hat{\alpha}, b) \leq W(g, 1, b)$; (ii) there exists a $\alpha_2 > \alpha_1$ such that $\bar{W}(\phi_0^*(\alpha_1), \phi_0^*(\alpha_2), g, \hat{\alpha}, b) = W(g, 1, b)$. Consider the former case. By Claim A.4, we have $\bar{W}(\phi_0^*(\alpha_1), \phi_0^*(\bar{\alpha}), g, \alpha, b) \leq W(g, 1, b)$ for all $\alpha \in (\alpha_1, \bar{\alpha}]$, which implies

$$W(g, \alpha, b) < \bar{W}(\phi_0^*(\alpha_1), \phi_0^*(\bar{\alpha}), g, \alpha, b) \leq W(g, 1, b)$$

for every $\alpha \in (\alpha_1, \bar{\alpha}]$, so the proof is done. In the latter case, we know $\phi_0^*(\alpha_1) \in [\underline{\phi}_0^*, \bar{\phi}_0^*)$ and $\phi_0^*(\alpha_2) \geq \bar{\phi}_0^*$. Hence, by Claim A.4, we have

$$\bar{W}(\phi_0^*(\alpha_1), \phi_0^*(\alpha_2), g, \alpha, b) < W(g, 1, b) = \bar{W}(\phi_0^*(\alpha_1), \phi_0^*(\alpha_2), g, \hat{\alpha}, b)$$

for all $\alpha \leq \bar{\alpha}$. Furthermore, since $\phi_0^*(\alpha) > \phi_0^*(\alpha_2)$ for all $\alpha \in (\alpha_1, \alpha_2)$, we have

$$W(g, \alpha, b) = \bar{W}(\phi_0^*(\alpha), \phi_0^*(\alpha), g, \alpha, b) < \bar{W}(\phi_0^*(\alpha_1), \phi_0^*(\alpha_2), g, \alpha, b) < W(g, 1, b)$$

for all $\alpha \in (\alpha_1, \alpha_2]$, where the first inequality follows from $\frac{\partial}{\partial \phi_\gamma} \bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b) > 0$ and $\frac{\partial}{\partial \phi_\delta} \bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b) < 0$. If $\alpha_2 \geq \bar{\alpha}$, then the proof is complete. If not, construct a sequence $\{\alpha_3, \alpha_4, \dots\}$ inductively such that $\bar{W}(\phi_0^*(\alpha_n), \phi_0^*(\alpha_{n+1}), g, \hat{\alpha}, b) = W(g, 1, b)$. Then, for every (α_n, α_{n+1}) in this sequence, we have

$$\bar{W}(\phi_0^*(\alpha_n), \phi_0^*(\alpha_n), g, \hat{\alpha}, b) < W(g, \hat{\alpha}, b) \leq 0 < \bar{W}(\phi_0^*(\alpha_n), \phi_0^*(\alpha_{n+1}), g, \hat{\alpha}, b) = W(g, 1, b),$$

where the first strict inequality follows from Claim A.3 and the second strict inequality holds by construction of the sequence (α_n) . This implies the sequence (α_n) should not have a limit point at some $\alpha' \leq \bar{\alpha}$. Hence, there must exist a $\alpha_N > \bar{\alpha}$ for some finite N , which implies $W(g, \alpha, b) < W(g, 1, b)$ for all $\alpha \leq \bar{\alpha}$.

We next prove for the remaining case $W(g, 1, b) \leq 0$. To this end, take an arbitrary positive number $\varepsilon \in (0, \bar{W}(g, \hat{\alpha}, b))$. Then, by applying the same logic above used for the case $W(g, 1, b) > 0$, one can find that $W(g, \alpha, b) < \varepsilon$ for every $\alpha \in (\hat{\alpha}, \bar{\alpha}]$. Since ε is an arbitrary number, we have that $W(g, \alpha, b) \leq 0$ for every $\alpha \in (\hat{\alpha}, \bar{\alpha}]$. Therefore, we find that $W(g, \alpha, b) < W(g, 1, b) \vee 0$ for every $\alpha \in (\hat{\alpha}, \bar{\alpha}]$, which completes the proof. $\square$

From Claim A.1 – A.5, we have that $W(g, \alpha, b) \leq W(g, 1, b) \vee 0$, where the equality holds if and only if $W(g, 1, b) > 0$ and $\alpha = 1$. This means $\alpha^{**}(b) \in \{0, 1\}$ for every $b > \frac{1-p}{x} - (1-g)$. $\square$

Lemma A.3. $\frac{d}{db} W(g, 1, b) > 0$ for every $b > \frac{1-p}{x} - (1-g)$.

Proof. For every b that yields $\phi_0^* < 1$ and $R_0^* = \frac{1}{x}$ when $\alpha = 1$, we have

$$W(g, 1, b) = [U(g, 1, b) - U(g, 0, b)] - \lambda \left[\left(\frac{p}{g} - x \right) - \left(\frac{p}{g} - \frac{1-p}{1-g} \right) \max\{1 - (1+b)x, 0\} \right].$$

Since $g > p$ by Lemma 3 and $\frac{d}{db} [U(g, 1, b) - U(g, 0, b)] > 0$ from (A.3), we have $\frac{d}{db} W(g, 1, b) > 0$, which is the desired result. $\square$

Proof of Lemma 5. By Lemma A.2, the optimal disclosure rule must be either $\alpha^{**}(b) = 0$ or $\alpha^{**}(b) = 1$ for each $b > \frac{1-p}{x} - (1-g)$. Since $\alpha^{**}(b)$ maximizes the Lagrangian, we must have $\alpha^{**}(b) = 1$ for $b > \frac{1-p}{x} - (1-g)$ if and only if $W(g, 1, b) > 0$. Moreover, since $W(g, 1, b)$ is increasing in b by Lemma A.3, there exists a unique $\bar{b}^{**} \geq \frac{1-p}{x} - (1-g)$ such that $W(g, 1, \bar{b}^{**}) = 0$. Then, the optimal disclosure policy must have $\alpha^{**}(b) = 0$ if $b \in \left(\frac{1-p}{x} - (1-g), \bar{b}^{**} \right)$ and $\alpha^{**}(b) = 1$ if $b \geq \bar{b}^{**}$. $\square$

Proof of Theorem 1. Fix an equilibrium fraction $g > p$ induced by an optimal disclosure policy $\alpha^{**}(\cdot)$. By Lemma 4, we have $\alpha^{**}(b) = 1$ if $b \leq \frac{g}{p} - 1$ and $\alpha^{**}(b) = 0$ if $\frac{g}{p} - 1 < b \leq \frac{1-p}{x} - (1-g)$. Furthermore, by Lemma 5, there exists a $\bar{b}^{**} \geq \frac{1-p}{x} - (1-g)$ such that $\alpha^{**}(b) = 0$ if $b \in \left(\frac{1-p}{x} - (1-g), \bar{b}^{**} \right)$ and $\alpha^{**}(b) = 1$ if $b > \bar{b}^{**}$. $\square$

A.3. Proofs for Section 5.1

In this section, we show that the main results of Lemma 5 and Theorem 1 are qualitatively unchanged if we modify our model by assuming that the date-2 utility function of each investor is represented by $u(x) = \frac{1}{1-\rho} x^{1-\rho}$ for some $\rho > 1$.

A.3.1. Date-1 equilibrium

Consider an individual investor at the capital market, who makes purchase offers (R_1^*, ϕ_1^*) to the banks labeled $s = 1$ and (R_0^*, ϕ_0^*) to the banks labeled $s = 0$, respectively. If the regulator reveals $s = 1$ with probability $\alpha \in [0, 1]$, the investor's expected utility is

$$\frac{p}{1-\rho} (y_G + \alpha \phi_1^* g R_1^* x + (1-\alpha) \phi_0^* g R_0^* x)^{1-\rho} + \frac{1-p}{1-\rho} (y_D + \phi_0^* (1-g) R_0^* x)^{1-\rho} - (\alpha \phi_1^* + (1-\alpha) \phi_0^*) g x - \phi_0^* ((1-g) + b)x,$$

where y_G and y_D are dividends equally distributed to every equity-holder. By differentiating with respect to ϕ_1^* , we have the following first order condition:

$$p \frac{\alpha g R_1^* x}{[(\alpha \phi_1^* + (1-\alpha) \phi_0^*) g]^\rho} - \alpha g x \geq 0.$$

Since $p - x > 0$, we have

$$(R_1^*, \phi_1^*) = \left(\frac{1}{p} [(\alpha \phi_1^* + (1-\alpha) \phi_0^*) g]^\rho, 1 \right) \text{ for all } b \geq 0.$$

Furthermore, by differentiating with respect to ϕ_0^* , we have the following first order condition:

$$\left[p \frac{(1-\alpha)g}{[(\alpha \phi_1^* + (1-\alpha) \phi_0^*) g]^\rho} + (1-p) \frac{(1-g)}{[(1-g) \phi_0^*]^\rho} \right] R_0^* \geq ((1-\alpha)g + (1-g) + b).$$

Therefore, we have

$$(R_0^*, \phi_0^*) = \left(\frac{(1-\alpha)g + (1-g) + b}{\frac{p}{g^{\rho-1}}(1-\alpha) + \frac{1-p}{(1-g)^{\rho-1}}}, 1 \right) \text{ if } \frac{(1-\alpha)g + (1-g) + b}{\frac{p}{g^{\rho-1}}(1-\alpha) + \frac{1-p}{(1-g)^{\rho-1}}} \leq \frac{1}{x}.$$

However, if $\frac{(1-\alpha)g + (1-g) + b}{\frac{p}{g^{\rho-1}}(1-\alpha) + \frac{1-p}{(1-g)^{\rho-1}}} > \frac{1}{x}$, and thus $R_0^* = \frac{1}{x}$, then ϕ_0^* is uniquely determined by the following equation:

$$\left[p \frac{(1-\alpha)g}{[(\alpha\phi_1^* + (1-\alpha)\phi_0^*)g]^\rho} + (1-p) \frac{(1-g)}{[(1-g)\phi_0^*]^\rho} \right] = ((1-\alpha)g + (1-g) + b)x.$$

By applying the Implicit Function Theorem, one can easily find that $\frac{d\phi_0^*}{d\alpha} < 0$ and $\frac{d\phi_0^*}{db} < 0$. From this result, we can construct the investors' expected utility function $U(g, \alpha, b)$ at $t = 1$ as follows:

$$U(g, \alpha, b) := \frac{p}{1-\rho} [(\alpha + (1-\alpha)\phi_0^*)g]^{1-\rho} - (\alpha + (1-\alpha)\phi_0^*)gx \\ + \frac{1-p}{1-\rho} [\phi_0^*(1-g)]^{1-\rho} - \phi_0^*((1-g) + b)x.$$

In particular, the expected utility function without the regulator's information is $U(g, 0, b)$, which is written as

$$U(g, 0, b) = \frac{p}{1-\rho} (\phi^*g)^{1-\rho} + \frac{1-p}{1-\rho} (\phi^*(1-g))^{1-\rho} - \phi^*(1+b)x,$$

$$\text{where } \phi^* = \left[\frac{pg^{1-\rho} + (1-p)(1-g)^{1-\rho}}{(1+b)x} \right]^{\frac{1}{\rho}} \wedge 1.$$

Lastly, the ex-post net gains from information disclosure is $U(g, \alpha, b) - U(g, 0, b)$. By applying the Envelope Theorem, we have

$$\frac{d}{d\alpha} [U(g, \alpha, b) - U(g, 0, b)] = (1 - \phi_0^*) \left[\frac{p}{[(\alpha + (1-\alpha)\phi_0^*)g]^\rho} - gx \right] \geq 0,$$

where the inequality strictly holds if and only if $\phi_0^* < 1$. That is, the regulator can improve the investors' ex-post welfare by revealing more information on banks' types to the market. Similarly, we have

$$\frac{d}{db} [U(g, \alpha, b) - U(g, 0, b)] = (\phi^* - \phi_0^*)x \geq 0, \quad (\text{A.7})$$

where the inequality is strict if and only if $\phi^* > \phi_0^*$, i.e., adverse selection is relatively severe.

A.3.2. Date-0 project choice

To analyze how the disclosure policy α influences banks' project choices, we next consider the date-0 equilibrium. Then, each bank will take into account the following equity-holders' expected surplus:

$$\frac{p}{1-\rho} [\alpha g(1 - R_1^*x) + (1-\alpha)g\phi_0^*(1 - R_0^*x) + z_G]^{1-\rho} + \frac{1-p}{1-\rho} [(1-g)\phi_0^*(1 - R_0^*x) + z_D]^{1-\rho} - C^*.$$

If a bank chooses project type γ rather than δ , the marginal surplus will be equal to

$$\Delta(g, \alpha, b) := p \frac{\alpha}{[(\alpha + (1-\alpha)\phi_0^*)g]^\rho} (1 - R_1^*x) \\ + \left[p \frac{1-\alpha}{[(\alpha + (1-\alpha)\phi_0^*)g]^\rho} - (1-p) \frac{1}{[(1-g)\phi_0^*]^\rho} \right] \phi_0^*(1 - R_0^*x).$$

Given a disclosure policy $\{\alpha(b)\}_{b \geq 0}$, the fraction g must be determined to make $\int \Delta(g, \alpha, b) dF(b) = 0$ in equilibrium.

If $\phi_0^* = 1$, we have

$$\Delta(g, \alpha, b) = \left(\frac{p}{g^\rho} - \frac{1-p}{(1-g)^\rho} \right) - \left[\alpha x + R_0^*x \left(\frac{p}{g^\rho}(1-\alpha) - \frac{1-p}{(1-g)^\rho} \right) \right]. \quad (\text{A.8})$$

By plugging $R_0^* = \frac{(1-\alpha)g + (1-g) + b}{\frac{p}{g^{\rho-1}}(1-\alpha) + \frac{1-p}{(1-g)^{\rho-1}}}$ into (A.8), $\Delta(g, \alpha, b)$ is rewritten as

$$\Delta(g, \alpha, b) = \left(\frac{p}{g^\rho} - \frac{1-p}{(1-g)^\rho} \right) - \left(1 + \frac{(1-g) + b}{g} \right) x + \left(1 + \frac{\frac{(1-g) + b}{g} - \frac{(1-p)g^{\rho-1}}{p(1-g)^{\rho-1}}}{(1-\alpha) + \frac{(1-p)g^{\rho-1}}{p(1-g)^{\rho-1}}} \right) \frac{(1-p)g^{1-\rho}}{p(1-g)^\rho} x. \quad (\text{A.9})$$

Hence, $\Delta(g, \alpha, b)$ is decreasing in α if and only if $\frac{(1-g) + b}{g} - \frac{(1-p)g^{\rho-1}}{p(1-g)^{\rho-1}} < 0$, or equivalently,

$$b < (1-g) \left[\frac{1-p}{p} \left(\frac{g}{1-g} \right)^\rho - 1 \right]. \quad (\text{A.10})$$

The right hand side of (A.10) is positive if and only if $g \in (0, 1)$ and $g^\rho > p$.

If $\phi_0^* < 1$, we have

$$\Delta(g, \alpha, b) = \alpha \left[\frac{p}{[(\alpha + (1-\alpha)\phi_0^*)g]^\rho} - x \right]. \quad (\text{A.11})$$

Lastly, since $\frac{d\phi^*}{d\alpha} < 0$, we can find $\hat{\alpha} \in [0, 1]$ for every $b \geq 0$ such that $\Delta(g, \alpha, b)$ is equal to (A.9) if $\alpha \leq \hat{\alpha}$ but equal to (A.11) if $\alpha > \hat{\alpha}$. Furthermore, by applying $\frac{d\phi^*}{d\alpha} < 0$, we have that (i) $\hat{\alpha} = 1$ for every $b \leq \frac{1-p}{x(1-g)^{\rho-1}} - (1-g)$; (ii) $\hat{\alpha} \in (0, 1)$ and $\hat{\alpha}$ is decreasing in b if $b \in \left(\frac{1-p}{x(1-g)^{\rho-1}} - (1-g), \frac{1}{x} \left[\frac{p}{g^{\rho-1}} + \frac{1-p}{(1-g)^{\rho-1}} \right] - 1 \right]$; (iii) $\hat{\alpha} = 0$ if $b \geq \frac{1}{x} \left[\frac{p}{g^{\rho-1}} + \frac{1-p}{(1-g)^{\rho-1}} \right] - 1$.

A.3.3. The ex-ante optimal disclosure policy

Proof of Theorem 2. First, consider $b \leq \frac{1-p}{x(1-g)^{\rho-1}} - (1-g)$. In this case, we have $\phi^* = \phi_0^* = \phi_1^* = 1$. Therefore, we have $U(g, \alpha, b) - U(g, 0, b) = 0$. Furthermore, we know from (A.9) that $\Delta(g, \alpha, b)$ is decreasing in α if and only if $\frac{(1-g)+b}{g} - \frac{(1-p)g^{\rho-1}}{p(1-g)^{\rho-1}} < 0$. Since the mathematical properties of $U(g, \alpha, b)$ and $\Delta(g, \alpha, b)$ are similar to those in the baseline model, we have the following lemmas, which correspond to Lemma 3 and 4, respectively.

Lemma A.4. Any optimal disclosure policy α^{**} must yield $g^\rho > p$.

Lemma A.5. For every $b \leq \frac{1-p}{x(1-g)^{\rho-1}} - (1-g)$, an optimal disclosure policy α^{**} is characterized by a threshold $\underline{b}^{**} := (1-g) \left[\frac{1-p}{p} \left(\frac{g}{1-g} \right)^\rho - 1 \right]$ such that $\alpha^{**}(b) = 1$ if $b \leq \underline{b}^{**}$ and $\alpha^{**}(b) = 0$ if $b \in \left(\underline{b}^{**}, \frac{1-p}{x(1-g)^{\rho-1}} - (1-g) \right]$.

Next, consider high $b > \frac{1-p}{x(1-g)^{\rho-1}} - (1-g)$. Like in the main model with $\rho = 1$, we know $U(g, \alpha, b) - U(g, 0, b) = 0$ for all $\alpha \leq \hat{\alpha}$, where $\hat{\alpha} > 0$. However, $\Delta(g, \alpha, b) - \Delta(g, 0, b) > 0$ since $\Delta(g, \alpha, b)$ is increasing in b for all $b > (1-g) \left[\frac{1-p}{p} \left(\frac{g}{1-g} \right)^\rho - 1 \right]$ and $\frac{1-p}{x(1-g)^{\rho-1}} - (1-g) > (1-g) \left[\frac{1-p}{p} \left(\frac{g}{1-g} \right)^\rho - 1 \right]$. These results imply that the regulator should not choose $\alpha \leq \hat{\alpha}$ for any b .

Lemma A.6. $\alpha^{**}(b) \geq \hat{\alpha}(b)$ for every $b > \frac{1-p}{x(1-g)^{\rho-1}} - (1-g)$ such that $\hat{\alpha}(b) > 0$.

Thus, like we did in the main model, consider the Lagrangian for a fixed $g > (p)^{\frac{1}{\rho}}$ corresponding to the regulator's ex-ante optimization problem:

$$\mathcal{L} := \int_{\frac{1-p}{x(1-g)^{\rho-1}} - (1-g)}^{\infty} [U(g, \alpha, b) - \lambda \Delta(g, \alpha, b)] dF(b),$$

where

$$\begin{aligned} [U - \lambda \Delta](g, \alpha, b) = & \left[\frac{p}{1-\rho} [(\alpha + (1-\alpha)\phi_0^*)g]^{1-\rho} - (\alpha + (1-\alpha)\phi_0^*)gx - \lambda \alpha \left(\frac{p}{[(\alpha + (1-\alpha)\phi_0^*)g]^\rho} - x \right) \right] \\ & + \left\{ \frac{1-p}{1-\rho} [\phi_0^*(1-g)]^{1-\rho} - \phi_0^*((1-g)+b)x \right\} \end{aligned} \quad (\text{A.12})$$

for every $\alpha \geq \hat{\alpha}$ and $b > \frac{1-p}{x(1-g)^{\rho-1}} - (1-g)$. Furthermore, define the net gains from information disclosure, denoted by $W(g, \alpha, b)$, as

$$\begin{aligned} W(g, \alpha, b) &:= [U(g, \alpha, b) - \lambda \Delta(g, \alpha, b)] - [U(g, 0, b) - \lambda \Delta(g, 0, b)] \\ &= [U(g, \alpha, b) - \lambda \Delta(g, \alpha, b)] - U(g, 0, b). \end{aligned}$$

From (A.12), we can define $\bar{W}(g, \alpha, b)$ and $\bar{W}(g, \alpha, b)$, modified forms of $W(g, \alpha, b)$, as follows:

$$\begin{aligned} \bar{W}(g, \alpha, b) &= \left[\frac{p}{1-\rho} [(\alpha + (1-\alpha)\bar{\phi}_0^*)g]^{1-\rho} - (\alpha + (1-\alpha)\bar{\phi}_0^*)gx - \lambda \alpha \left(\frac{p}{[(\alpha + (1-\alpha)\bar{\phi}_0^*)g]^\rho} - x \right) \right] \\ &+ \left\{ \frac{1-p}{1-\rho} [\bar{\phi}_0^*(1-g)]^{1-\rho} - \bar{\phi}_0^*((1-g)+b)x \right\} - U(g, 0, b), \end{aligned}$$

and

$$\begin{aligned} \bar{W}(\phi_\gamma, \phi_\delta, g, \alpha, b) &= \left[\frac{p}{1-\rho} [(\alpha + (1-\alpha)\phi_\gamma)g]^{1-\rho} - (\alpha + (1-\alpha)\phi_\gamma)gx - \lambda \alpha \left(\frac{p}{[(\alpha + (1-\alpha)\phi_\gamma)g]^\rho} - x \right) \right] \\ &+ \left\{ \frac{1-p}{1-\rho} [\phi_\delta(1-g)]^{1-\rho} - \phi_\delta((1-g)+b)x \right\} - U(g, 0, b); \end{aligned}$$

One can find that Claim A.1 – A.5 hold true, which then implies that Lemma A.2 holds true, too. Furthermore, $W(g, 1, b)$ is increasing in b by (A.7), which implies that Lemma A.3 holds true. By combining all the results together, we have that the optimal disclosure policy $\alpha^{**}(b)$ is characterized by two cutoffs $0 < \underline{b}^{**} < \bar{b}^{**} < \infty$ such that $\alpha^{**}(b) = 1$ if $b \in [0, \underline{b}^{**}] \cup [\bar{b}^{**}, \infty)$ and $\alpha^{**}(b) = 0$ otherwise. $\square$

A.4. Proofs for Section 5.3

Proof of Theorem 3. Fix any $b \geq 0$. If the regulator chooses no disclosure $\alpha = 0$, g is determined by

$$\Delta(g, 0, b) = \left(\frac{p}{g} - \frac{1-p}{1-g} \right) \max\{1 - (1+b)x, 0\} = 0.$$

If $b < \frac{1}{x} - 1$, no disclosure yields $g = p$ and the total welfare $U(p, 0, b)$. If $b \geq \frac{1}{x} - 1$, no disclosure yields any $g \in (0, 1)$ and the total welfare $U(g, 0, b)$. If the regulator chooses $\alpha > 0$, there are two possible cases, either $R_0^* < \frac{1}{x}$ or $R_0^* = \frac{1}{x}$. First, suppose the regulator's policy $\alpha > 0$ yields $R_0^* < \frac{1}{x}$. Then we have

$$\Delta(g, \alpha, b) = \left[\left(\frac{p}{g} - \frac{1-p}{1-g} \right) - \left(1 + \frac{(1-g)+b}{g} \right) x \right] + \frac{x(1-p)}{p(1-g)} \left(1 + \frac{\frac{(1-g)+b}{g} - \frac{1-p}{p}}{(1-\alpha) + \frac{1-p}{p}} \right).$$

If $\Delta(p, \alpha, b) < 0$, the disclosure policy $\alpha > 0$ yields $g \neq p$ and the resulting total welfare $U(g, \alpha, b) = U(g, 0, b)$, where the equality follows from $\phi_1^* = \phi_0^* = \phi^* = 1$ when $R_0^* < \frac{1}{x}$. If $\Delta(p, \alpha, b) = 0$, the disclosure policy yields $g = p$ and the resulting total welfare $U(p, \alpha, b) = U(p, 0, b)$. Since $\max_{g \in [0,1]} U(g, 0, b) = U(p, 0, b)$, it is weakly optimal for the regulator to choose no disclosure ($\alpha = 0$).

Next, suppose $\alpha > 0$ yields $R_0^* = \frac{1}{x}$. Then, we have

$$\Delta(g, \alpha, b) = \alpha \left(\frac{p}{(\alpha + (1-\alpha)\phi_0^*)g} - x \right) > 0,$$

where the strict inequality follows from $\alpha > 0$, $p > x$, and $g \in [0, 1]$. Therefore, any disclosure $\alpha > 0$ yields $g = 1$ and the resulting welfare $U(1, \alpha, b) = -\infty$. Therefore, it is optimal for the regulator to choose $\alpha = 0$. To sum up, a disclosure policy $\alpha^*(b) = 0$ for every $b \geq 0$ is weakly optimal. $\square$

Appendix. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfineco.2023.103760>.

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